Network Centrality in the European Competition Policy

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AEDE, Lisbon
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Crash course in (sectoral) network theory

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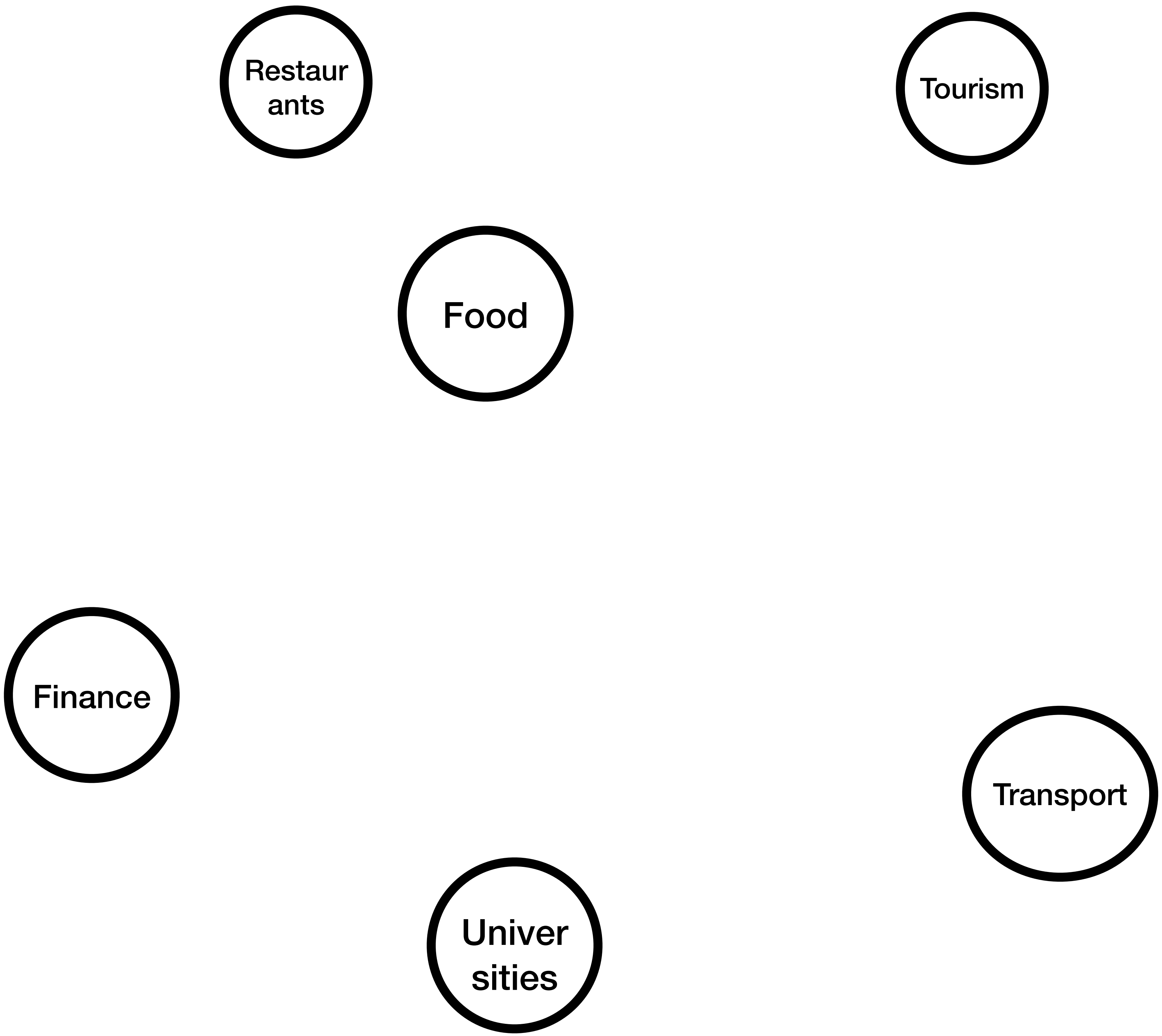

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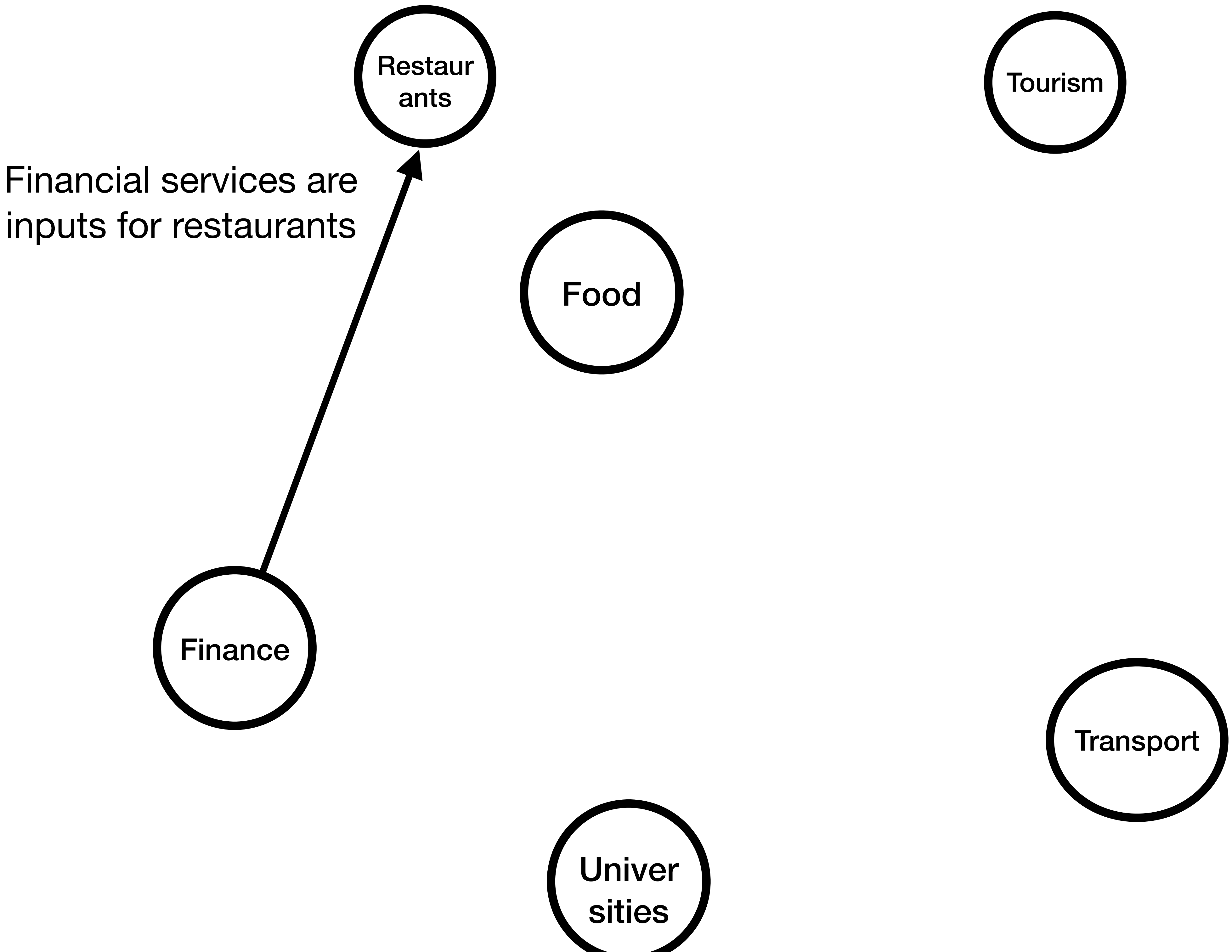

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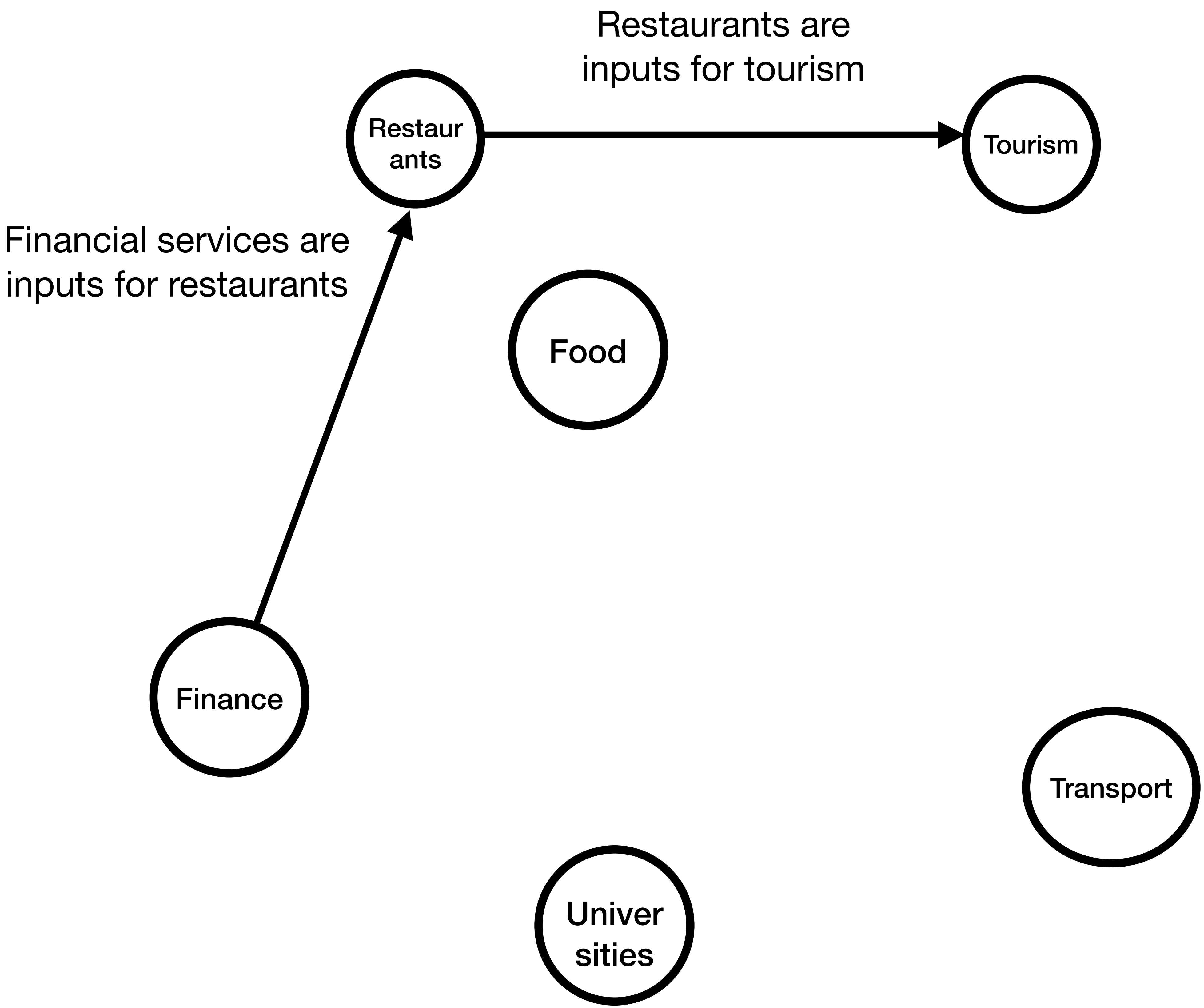

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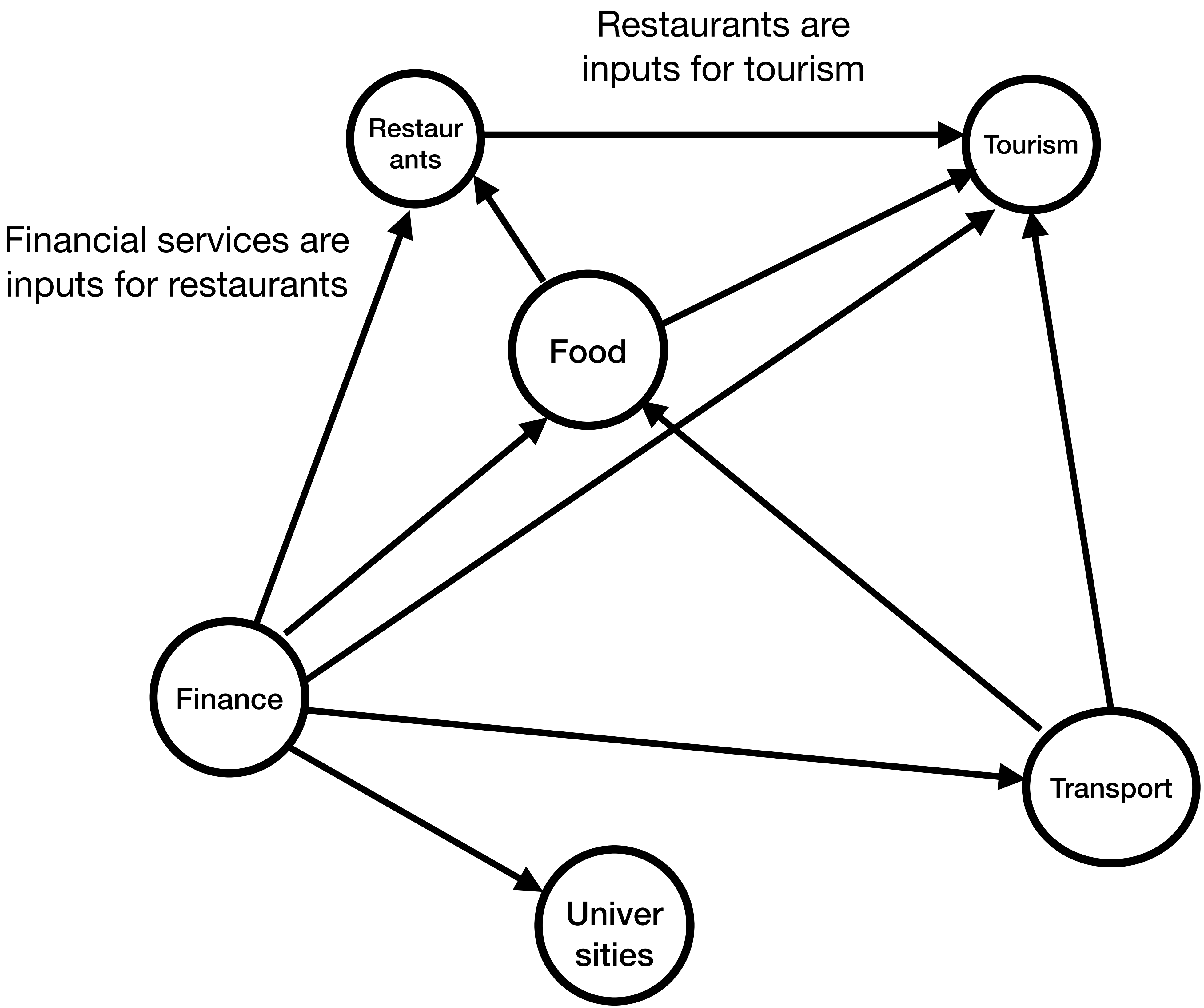

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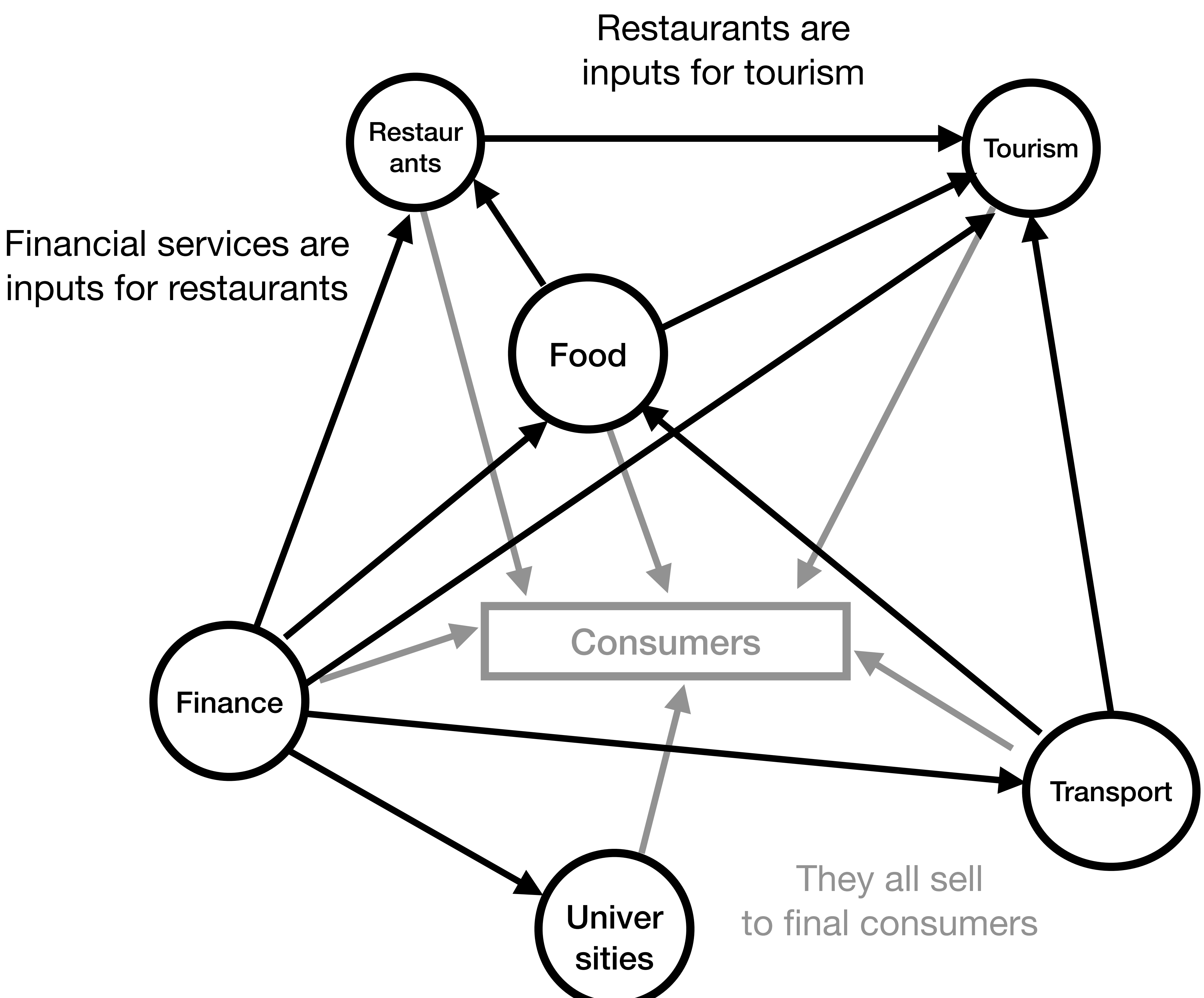

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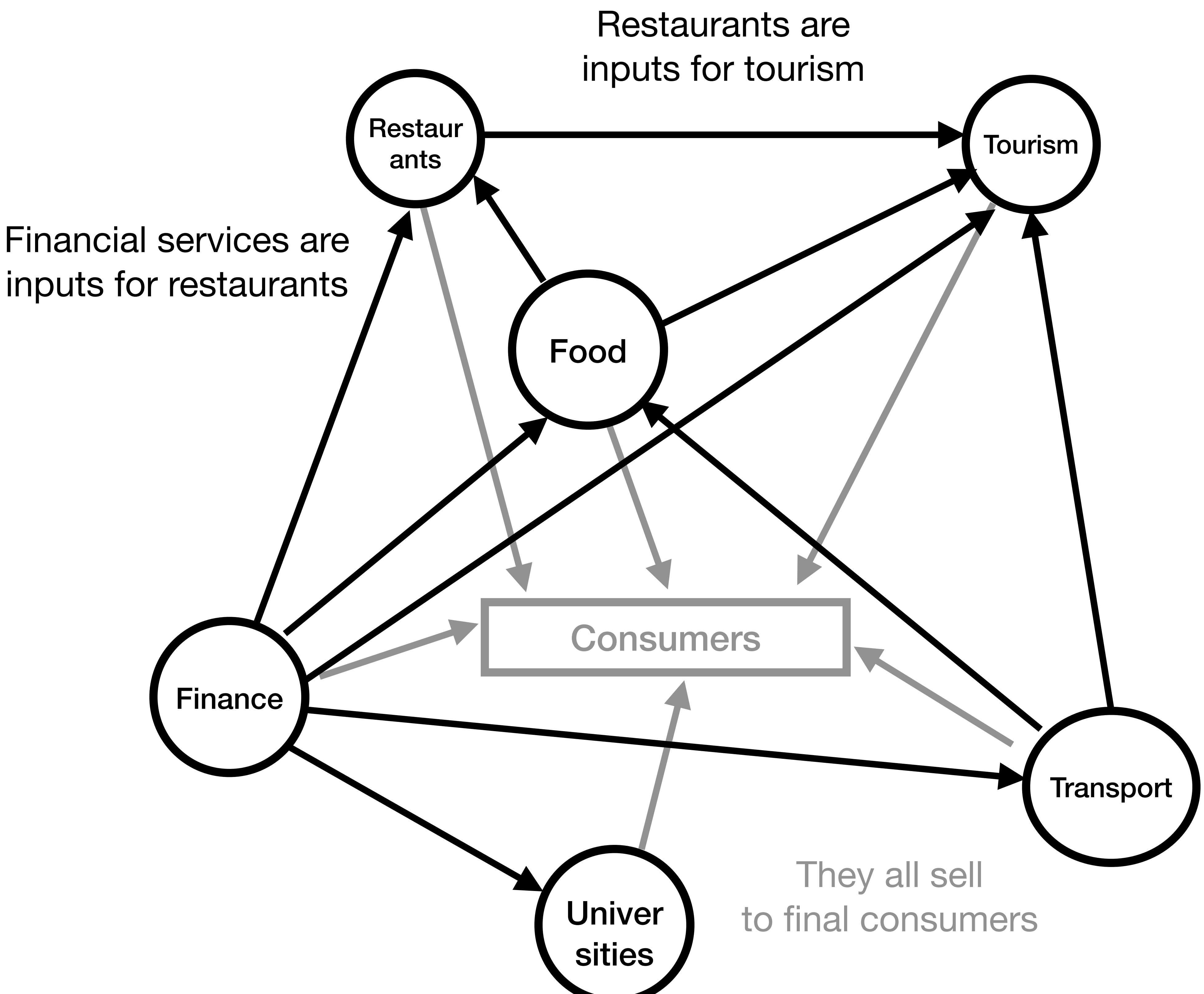

What's the most central sector?

Central = Important

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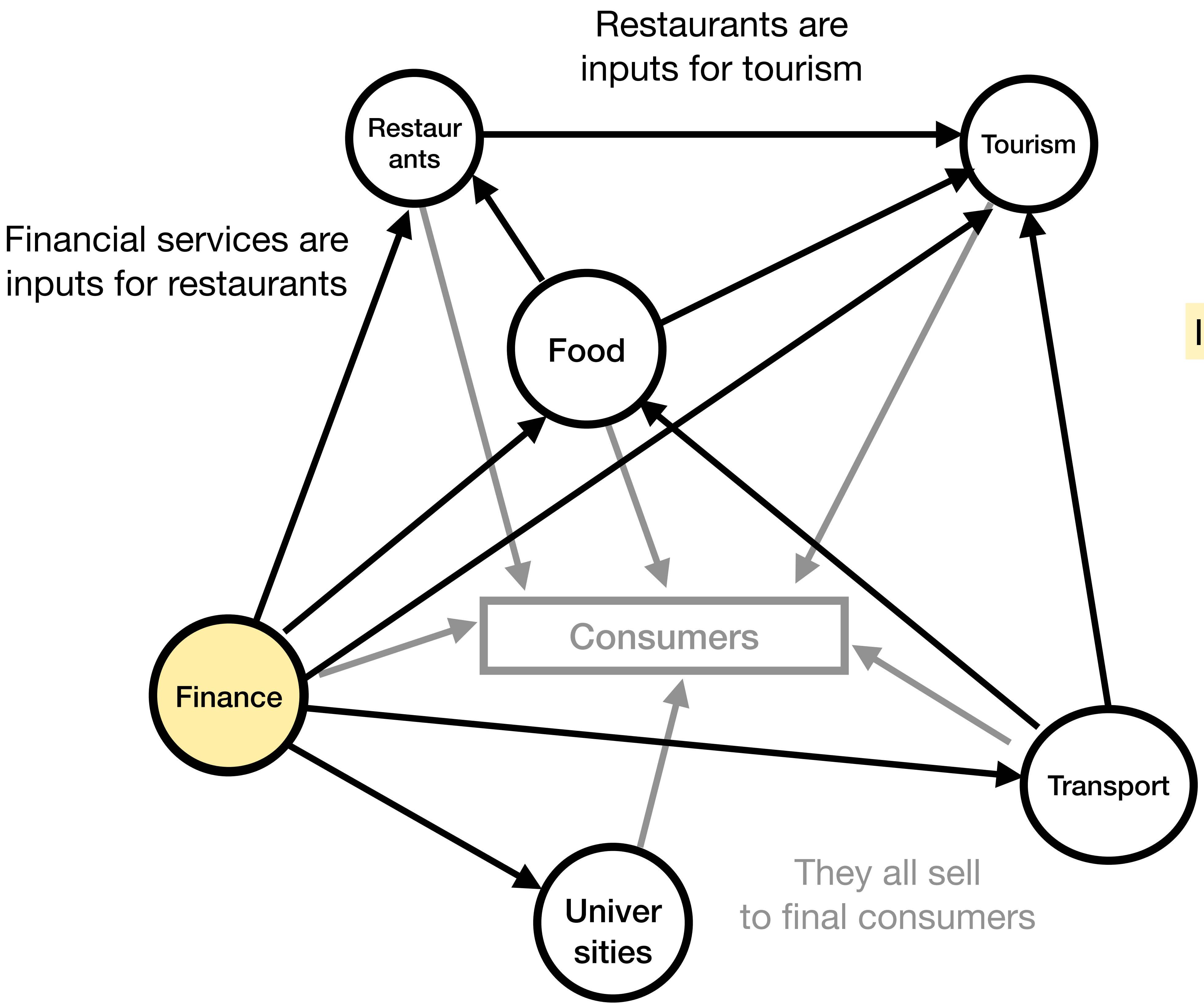

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Input production: **Finance**

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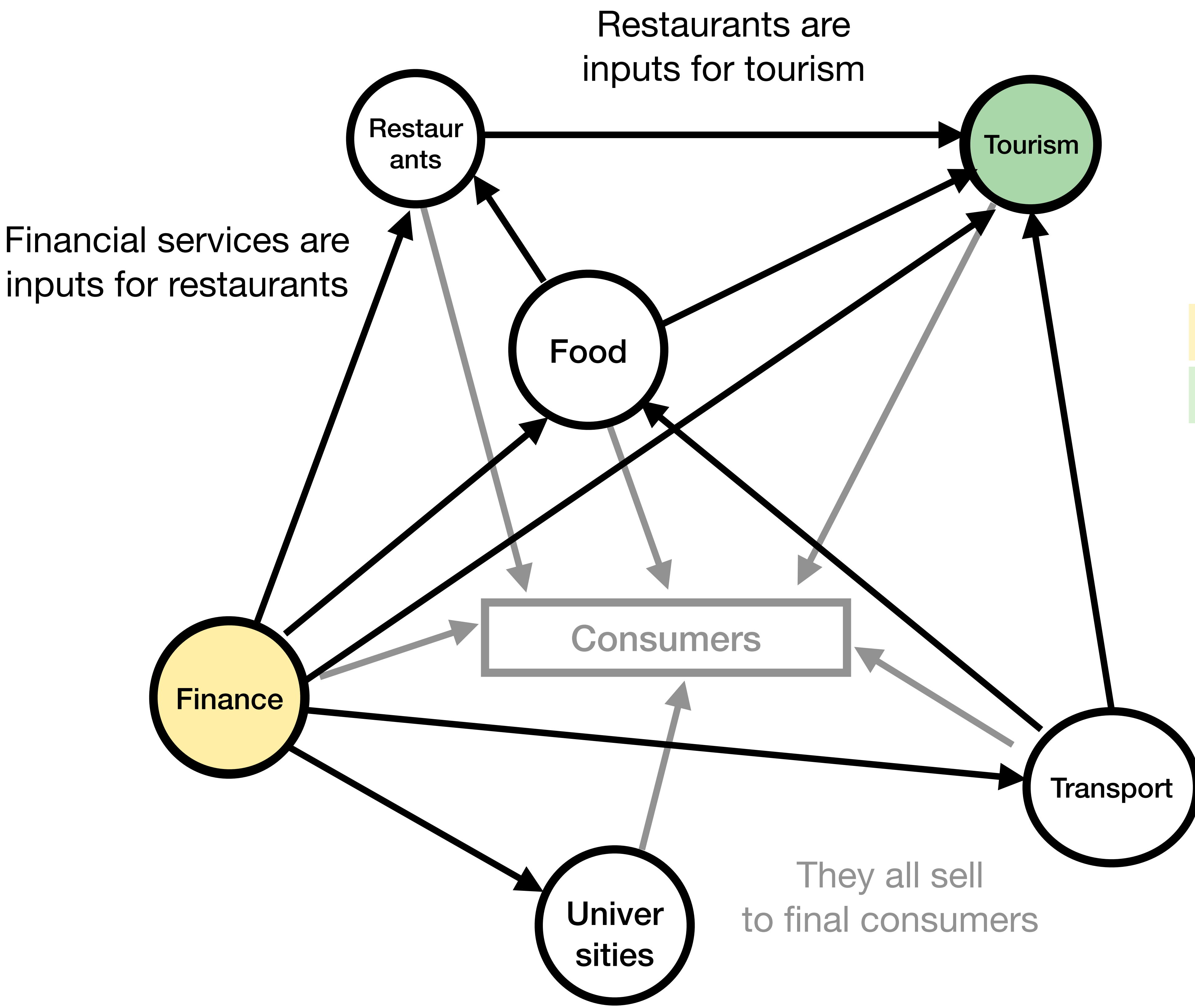

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Input production: **Finance**

Consumption: Tourism

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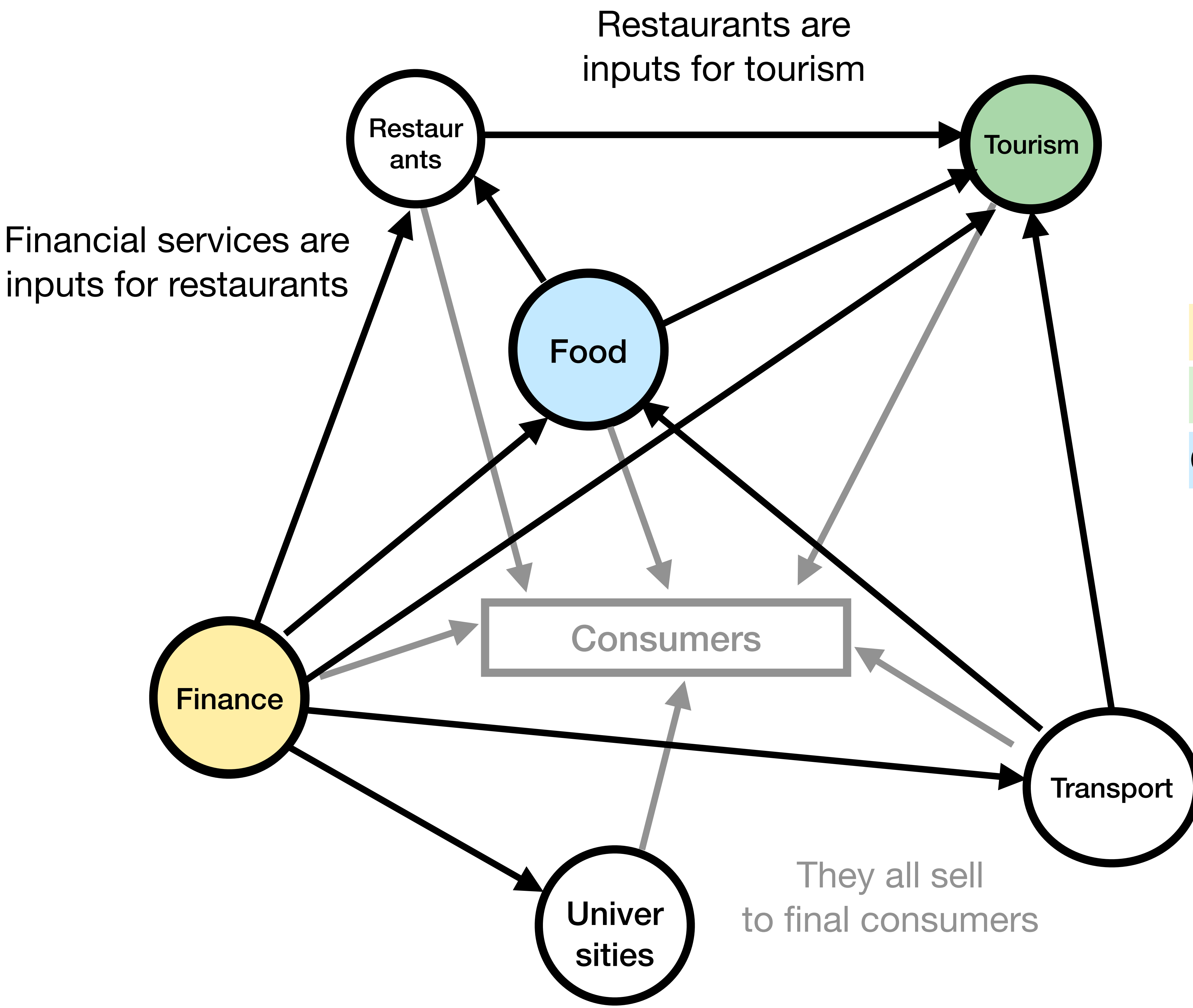

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Considering both:	Food

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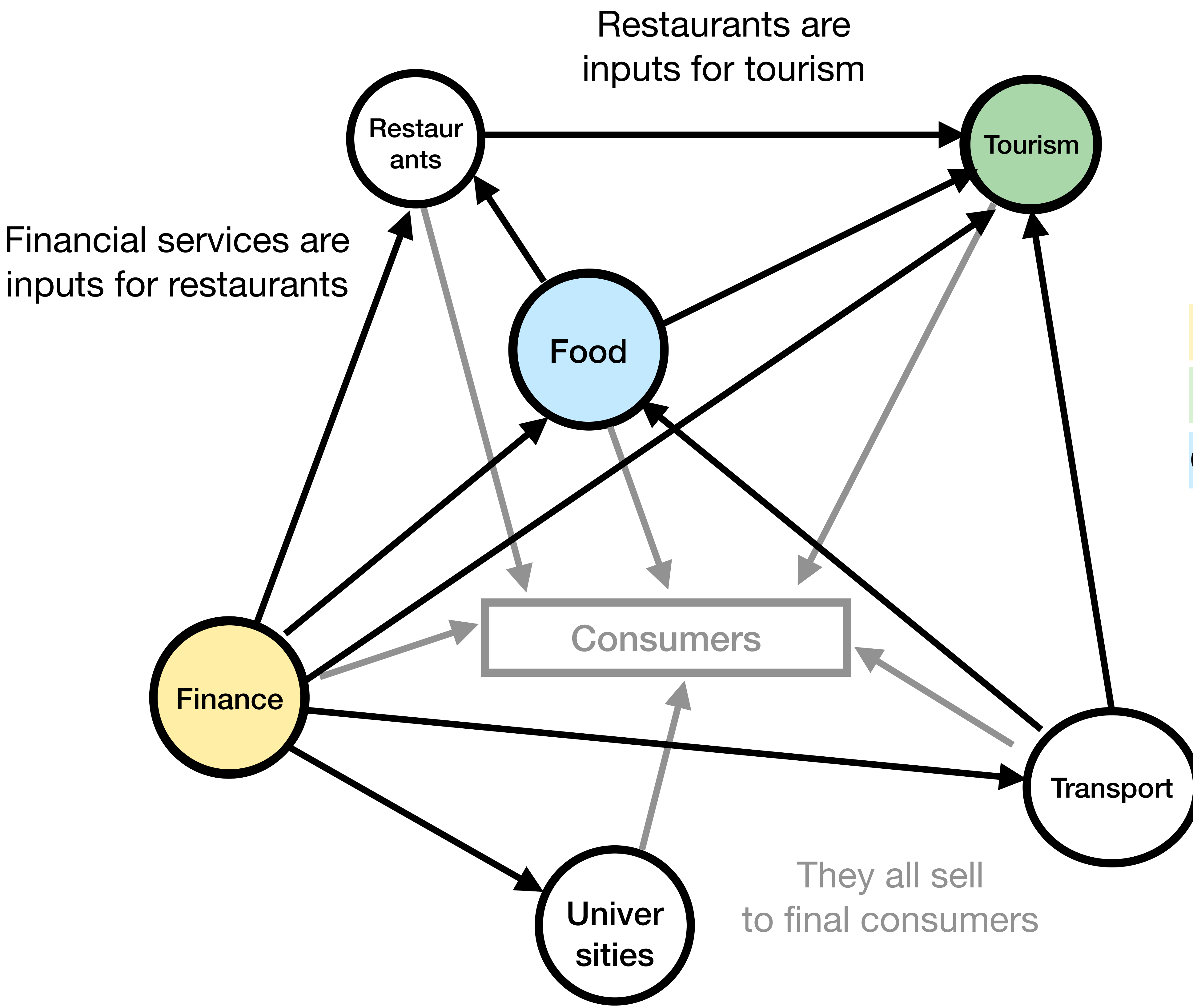

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Shocks (eg crisis) in central sectors more important to the GDP

(Baqaee and Farhi)

Motivation

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Why do we care? Brief answer: politicians care (Draghi, Ribera…)

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Today: *"2 papers in 1"*

1. Theoretically: How effects of a merger depend on centrality
2. Empirically: does the EU Commission take centrality into account?

Findings

Theory:

1. Merging parties in central sectors have **less incentives to act anticompetitively**

2. **Larger effects** (positive or negative) on the aggregate economy

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From 1990 to 2014, EU Commission cases:

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Question for you: **Why?** (We'll discuss later)

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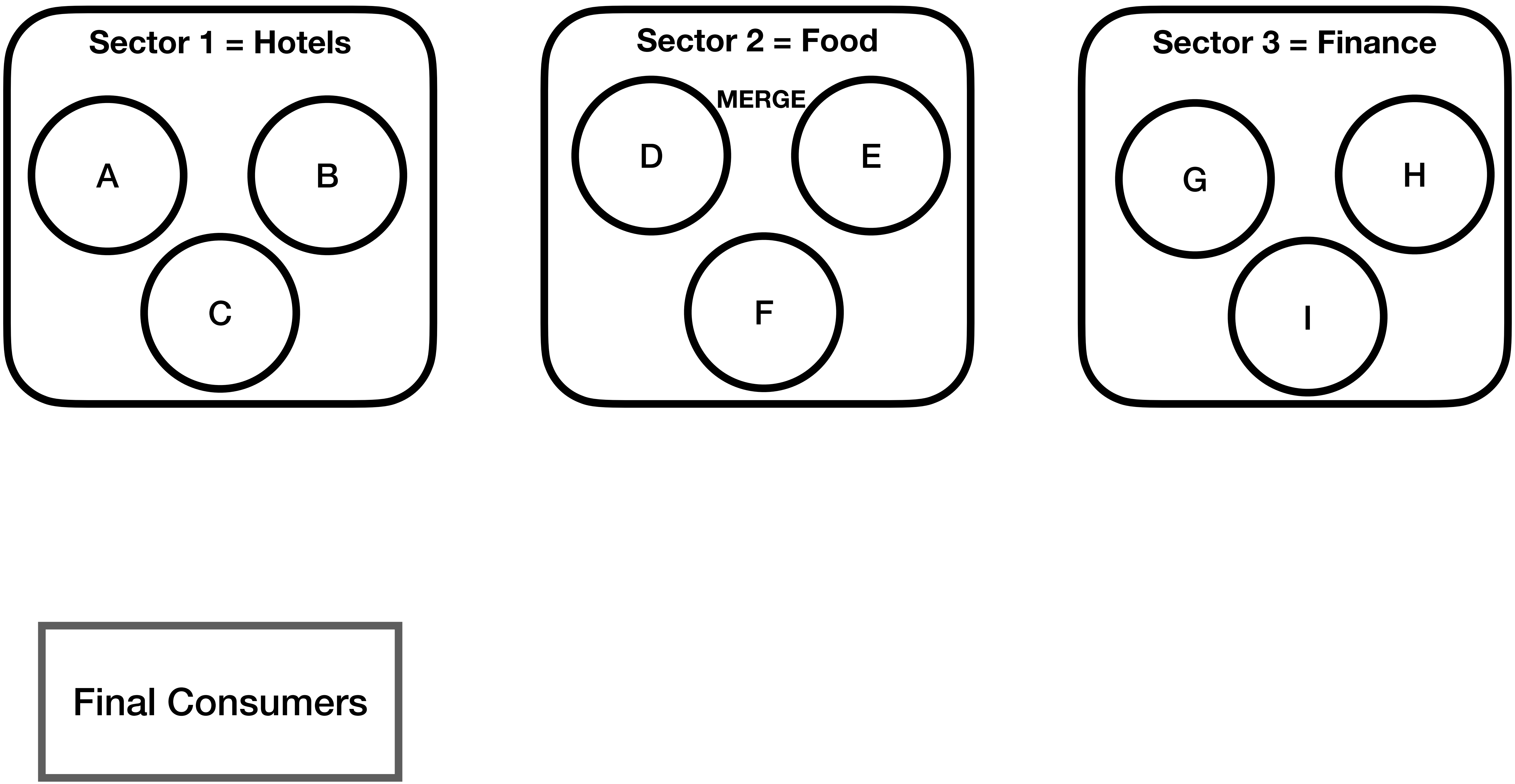
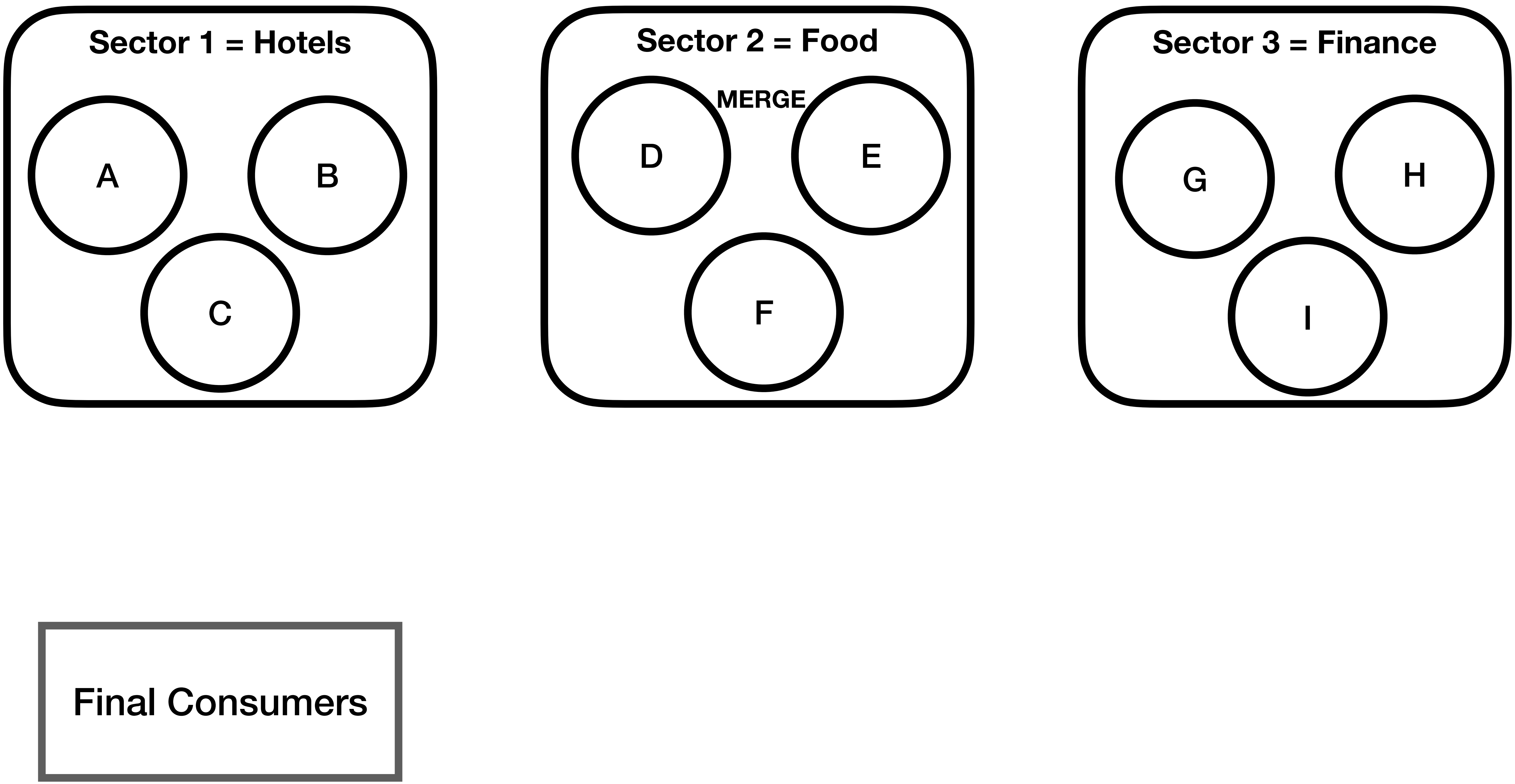


Theoretical Model (omitted + simplified)

Goods are homogeneous within sectors, Cournot competition

Complementary or substitutes across sectors



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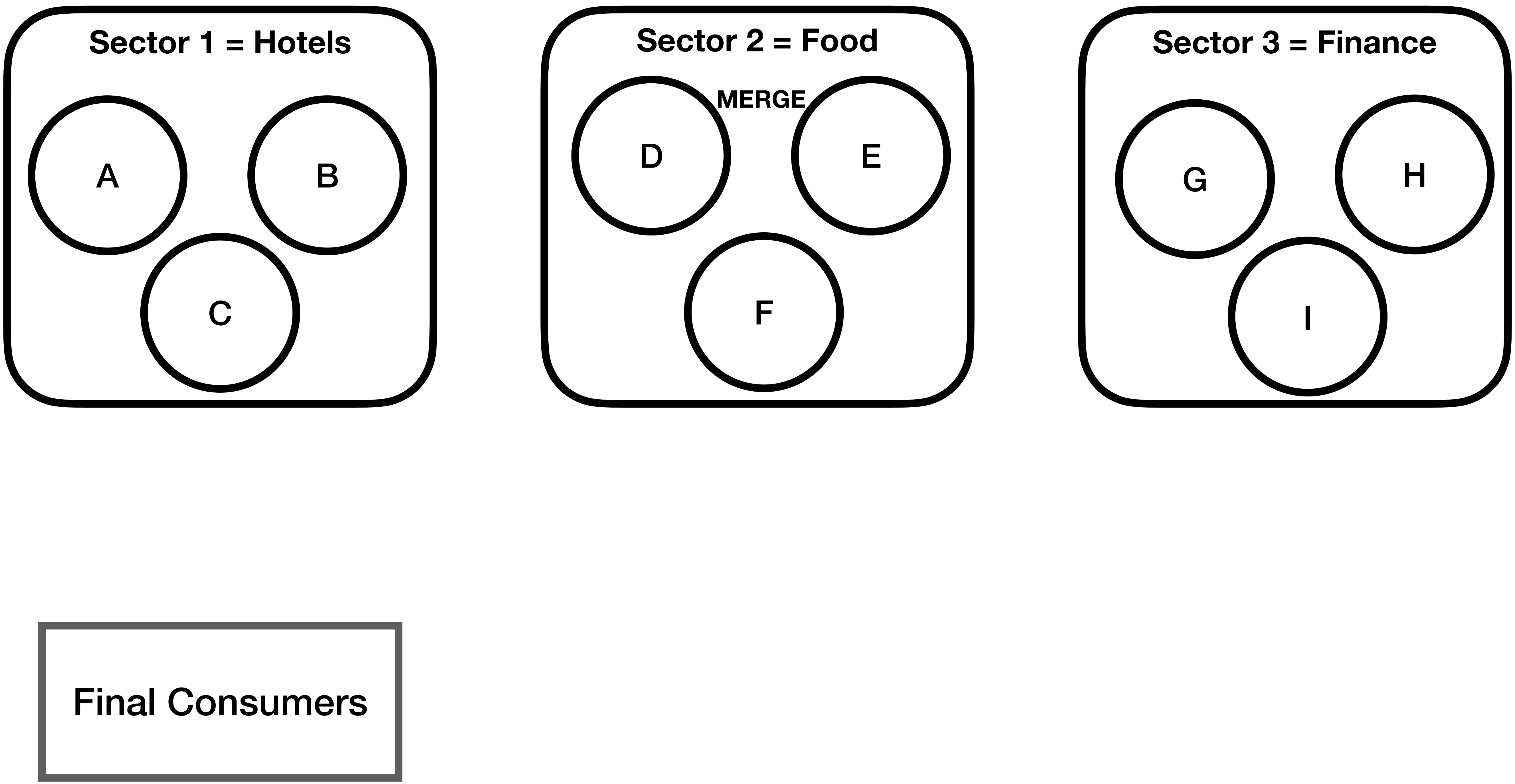

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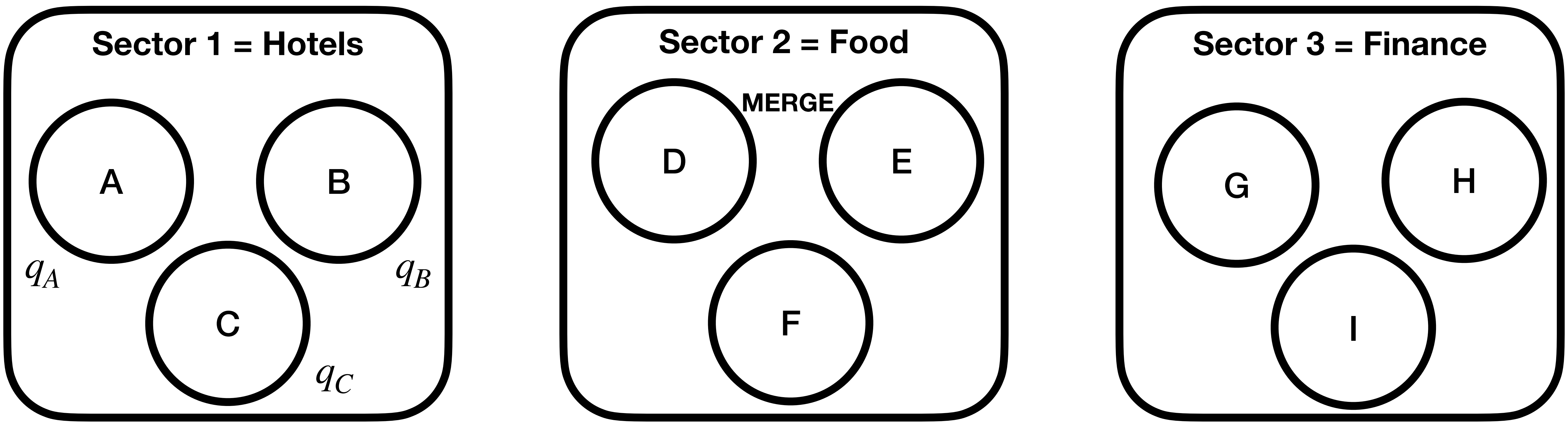

Final Consumers

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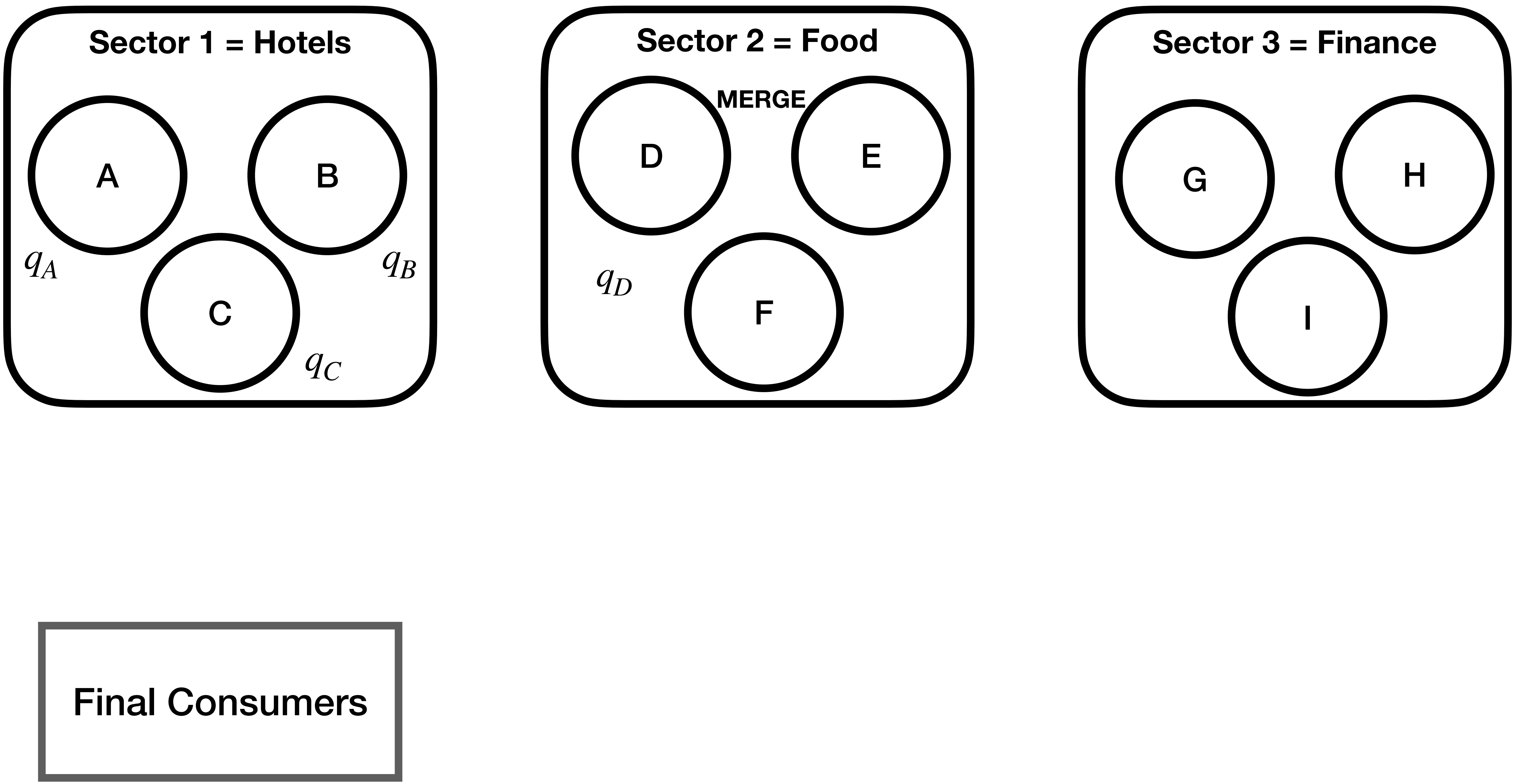

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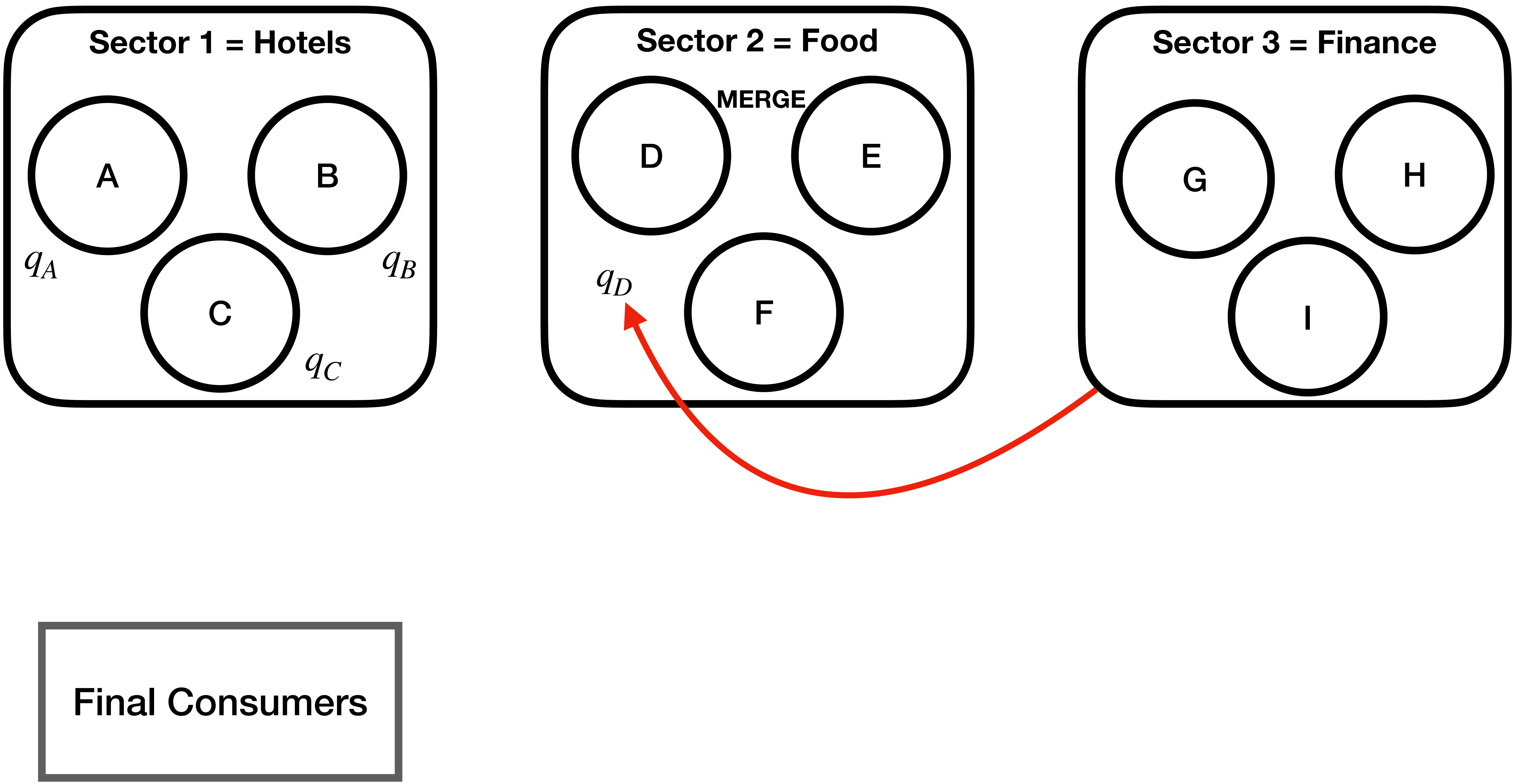

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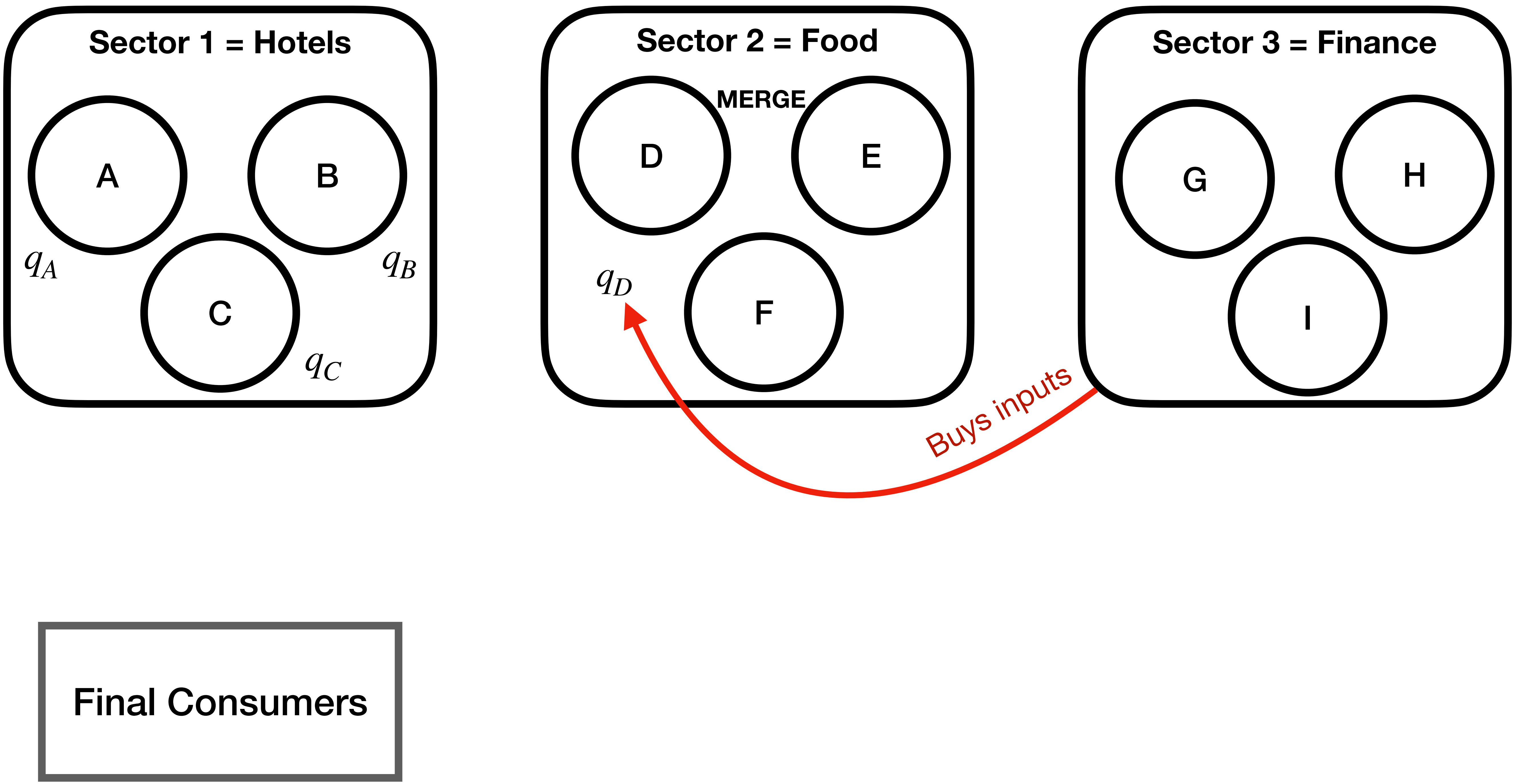

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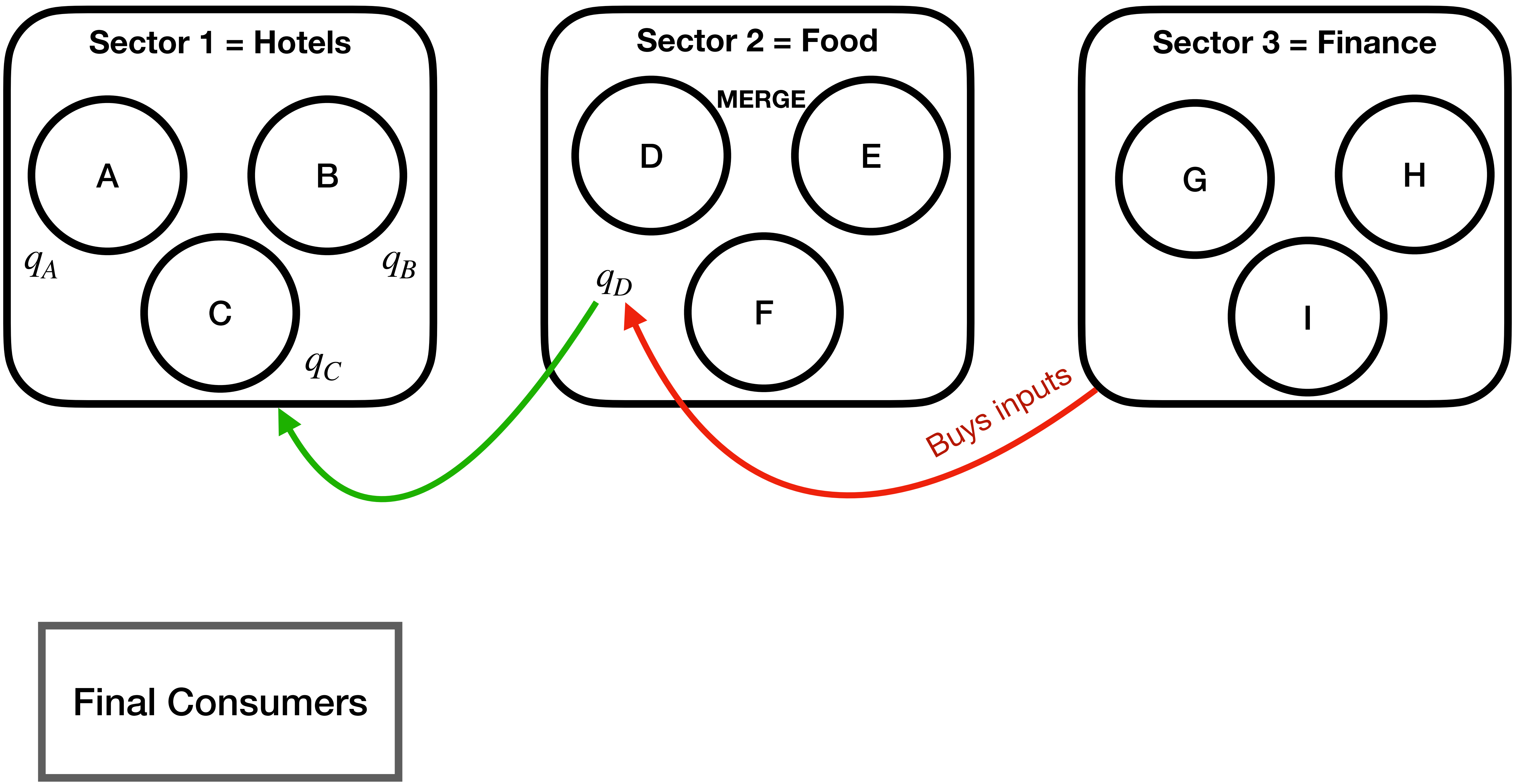

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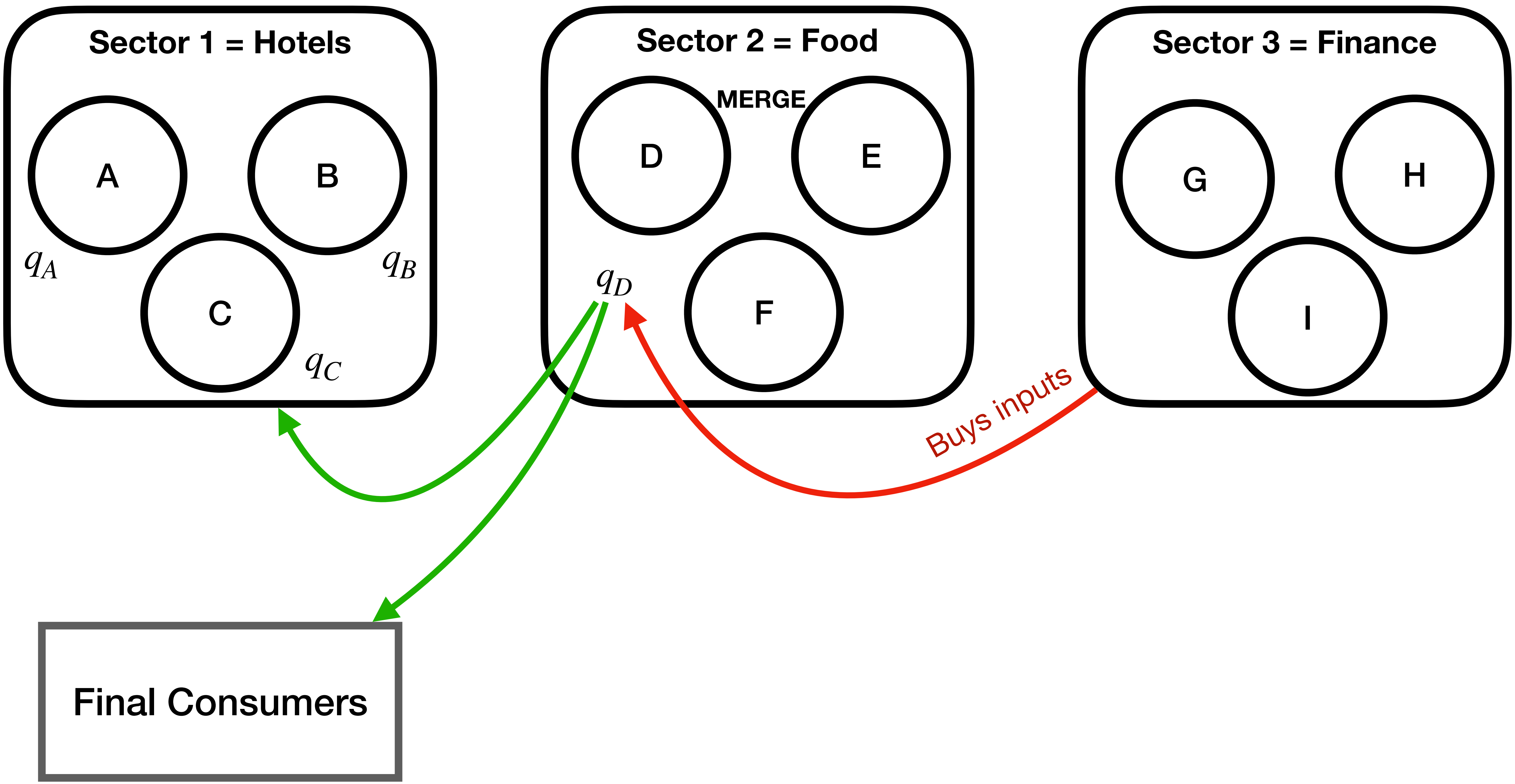

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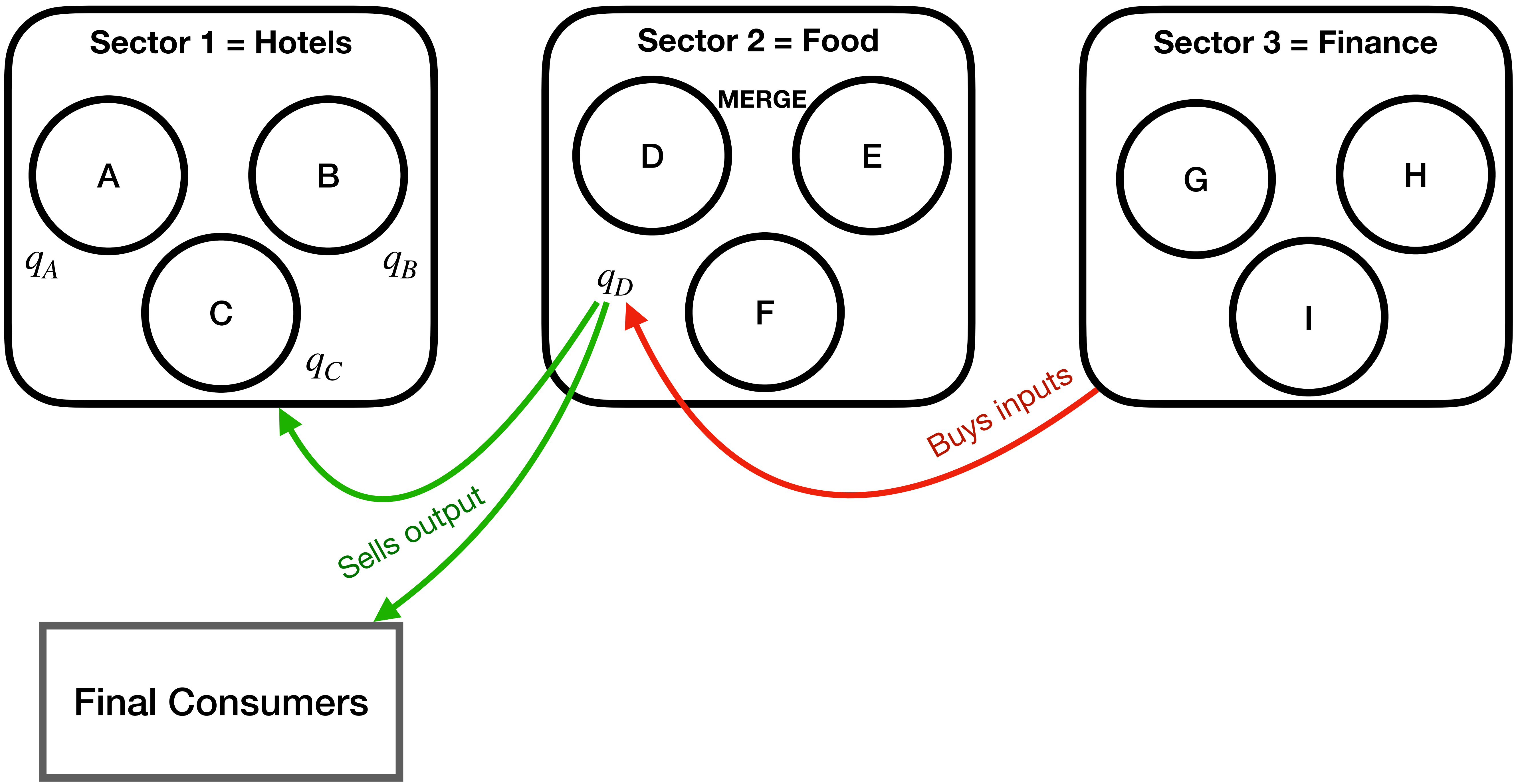

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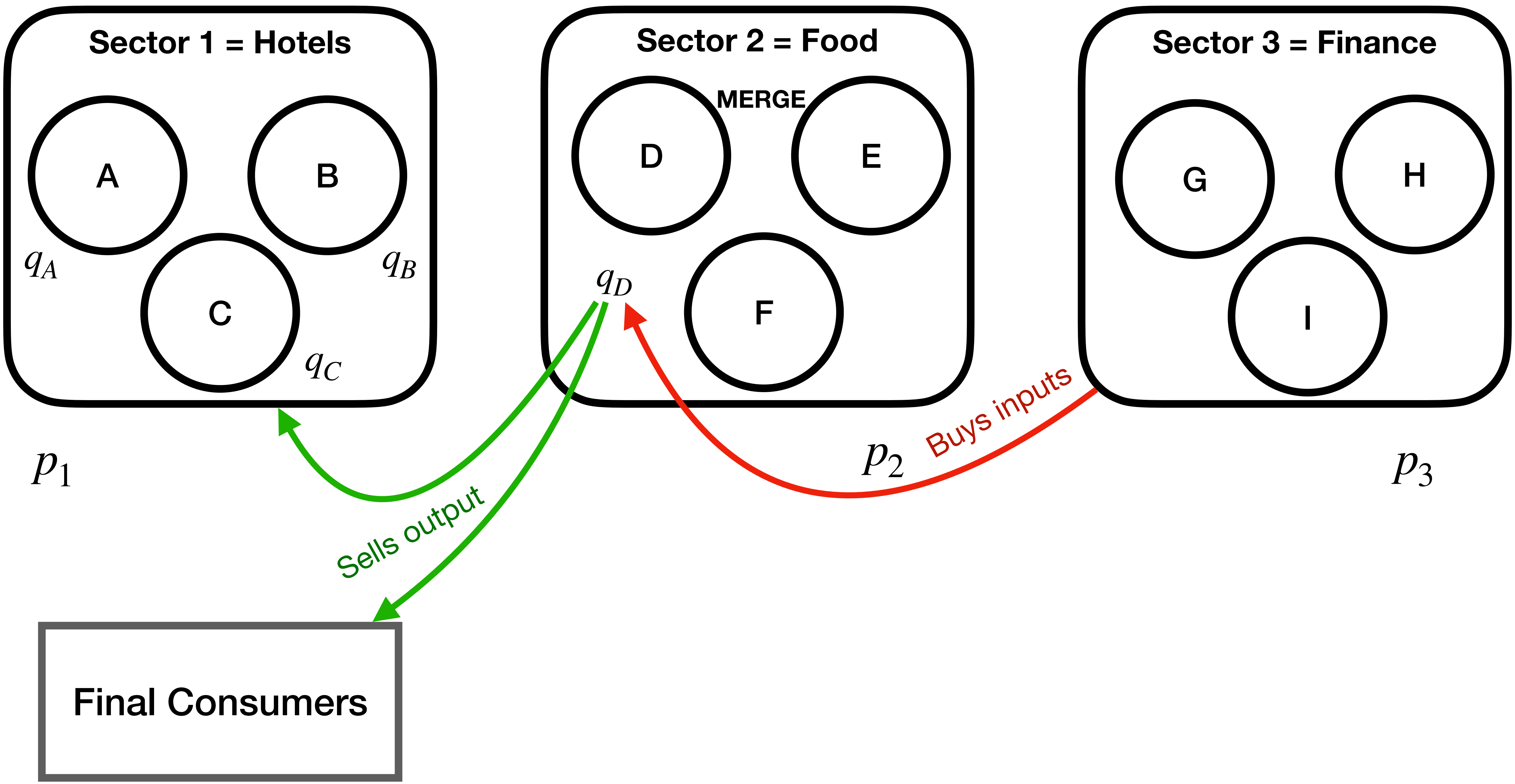

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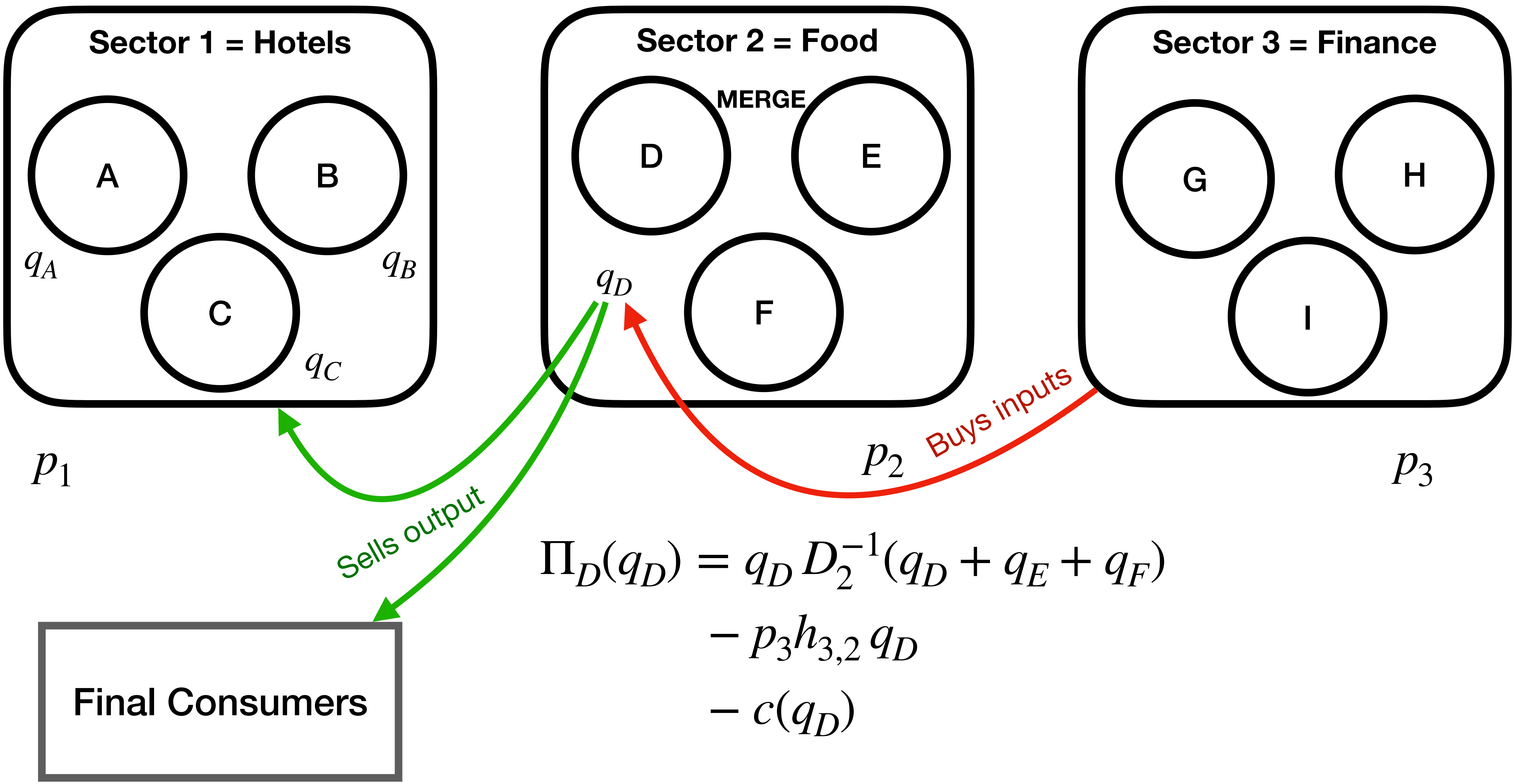

$$\Pi_D(q_D) = q_D\, D_2^{-1}(q_D + q_E + q_F)$$
$$- p_3 h_{3,2}\, q_D$$
$$- c(q_D)$$

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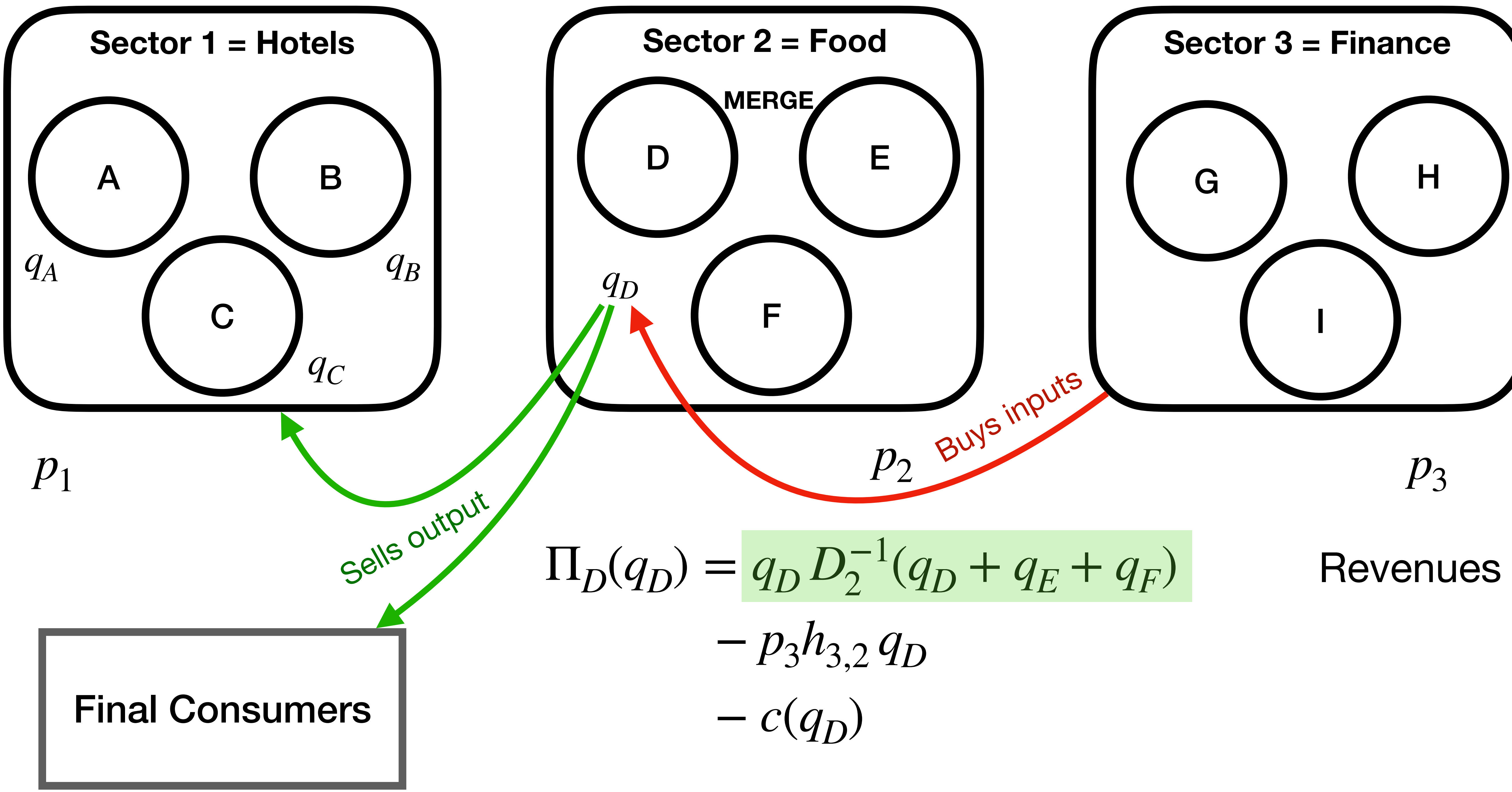

$$\Pi_D(q_D) = q_D\, D_2^{-1}(q_D + q_E + q_F)$$ Revenues

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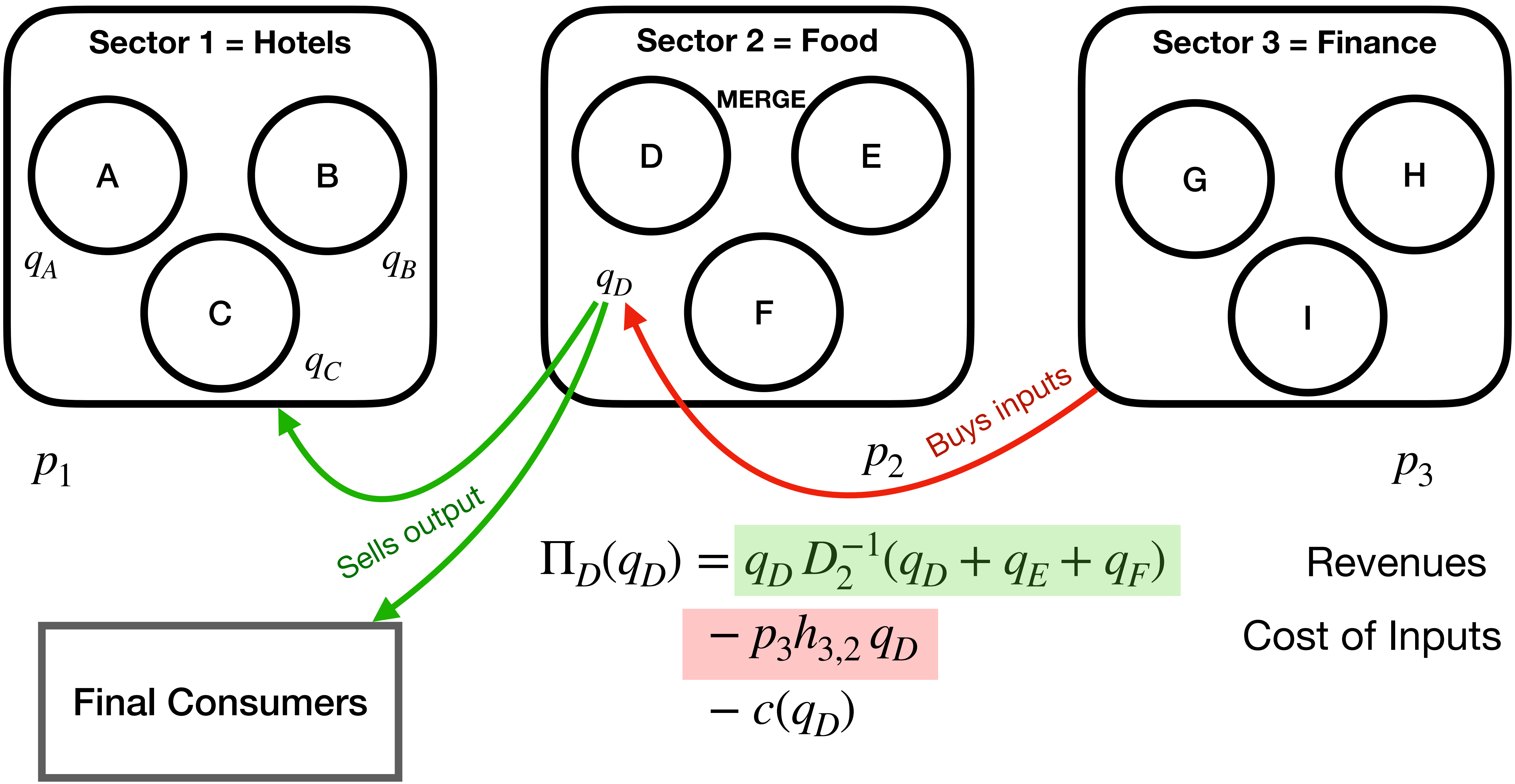

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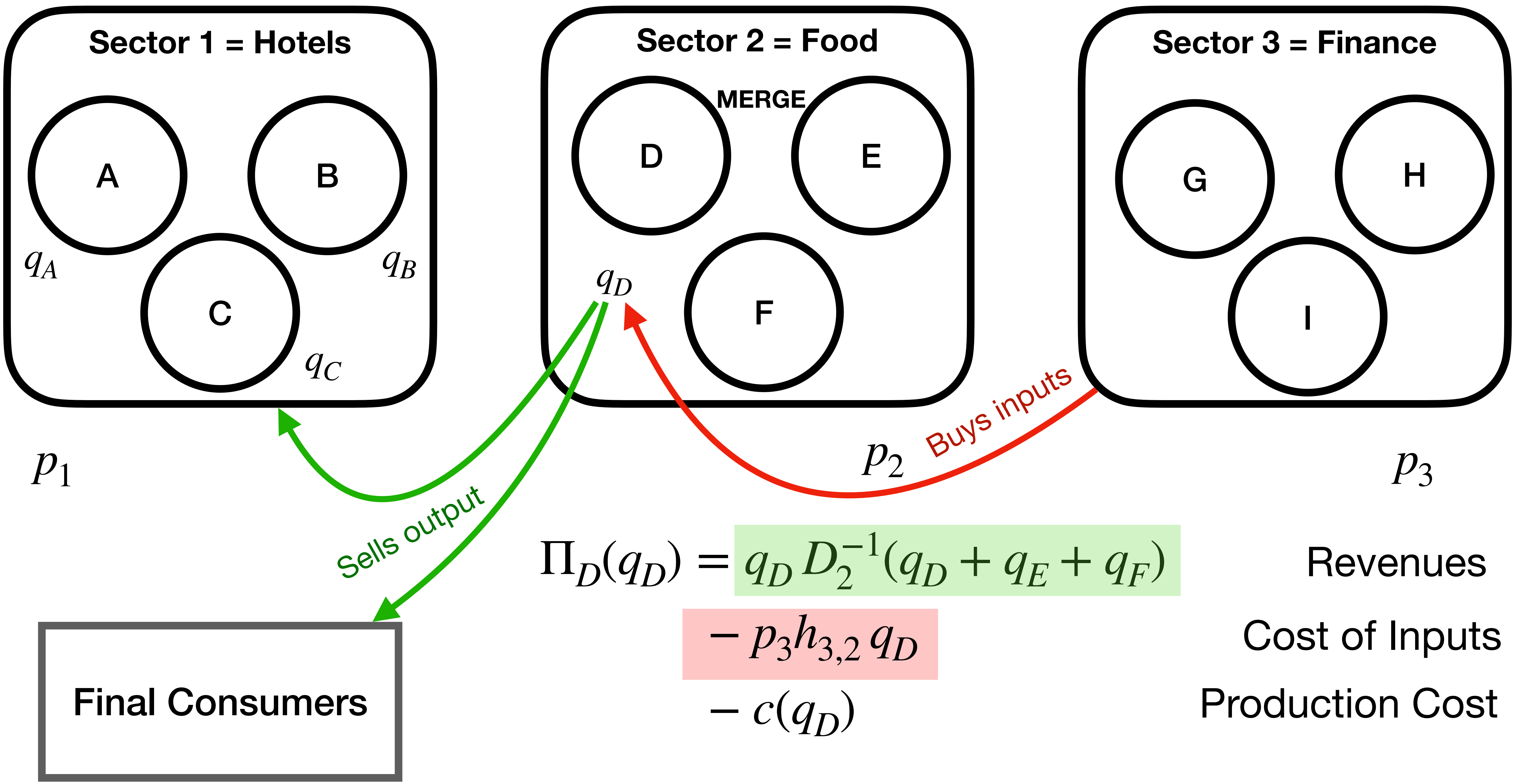

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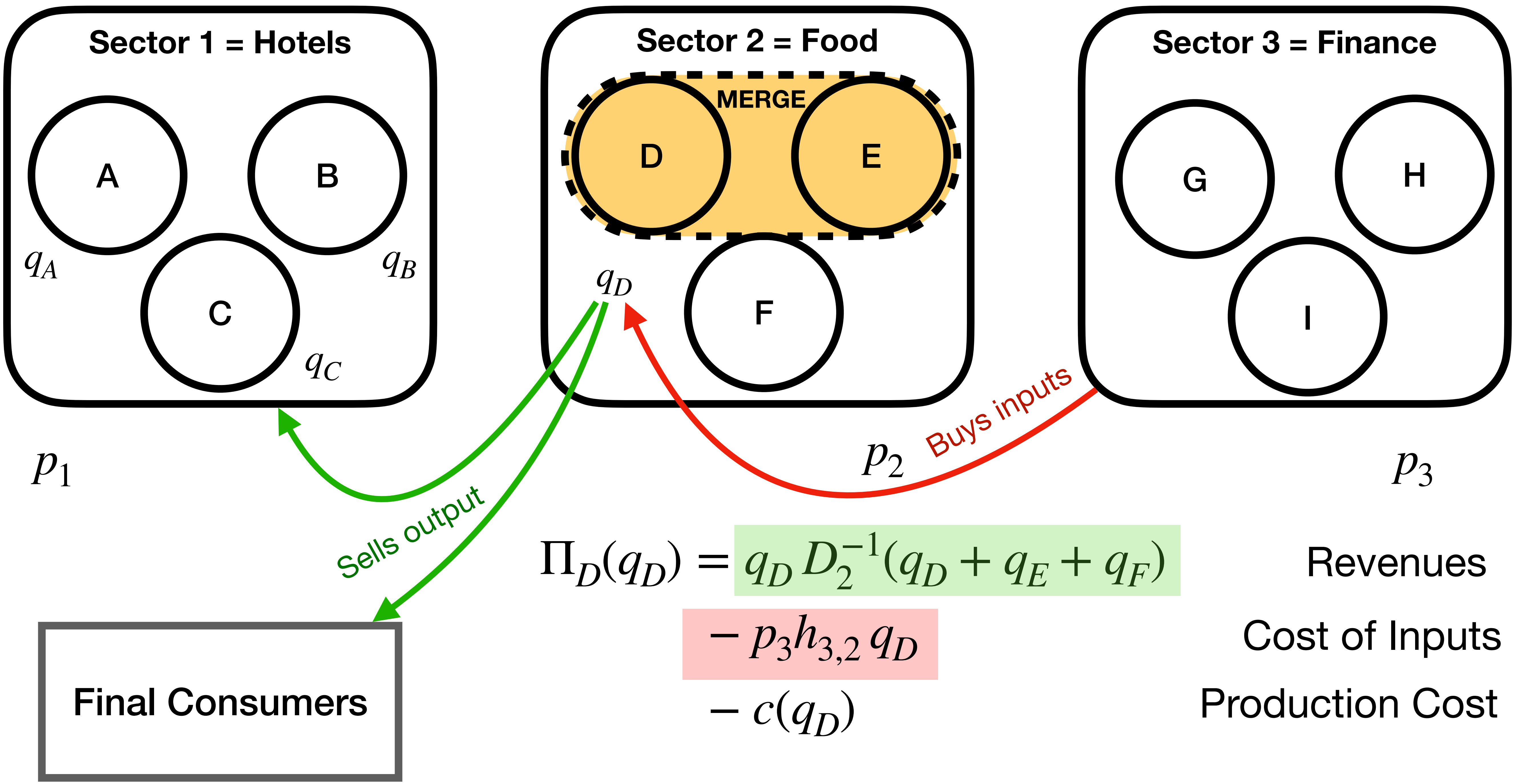

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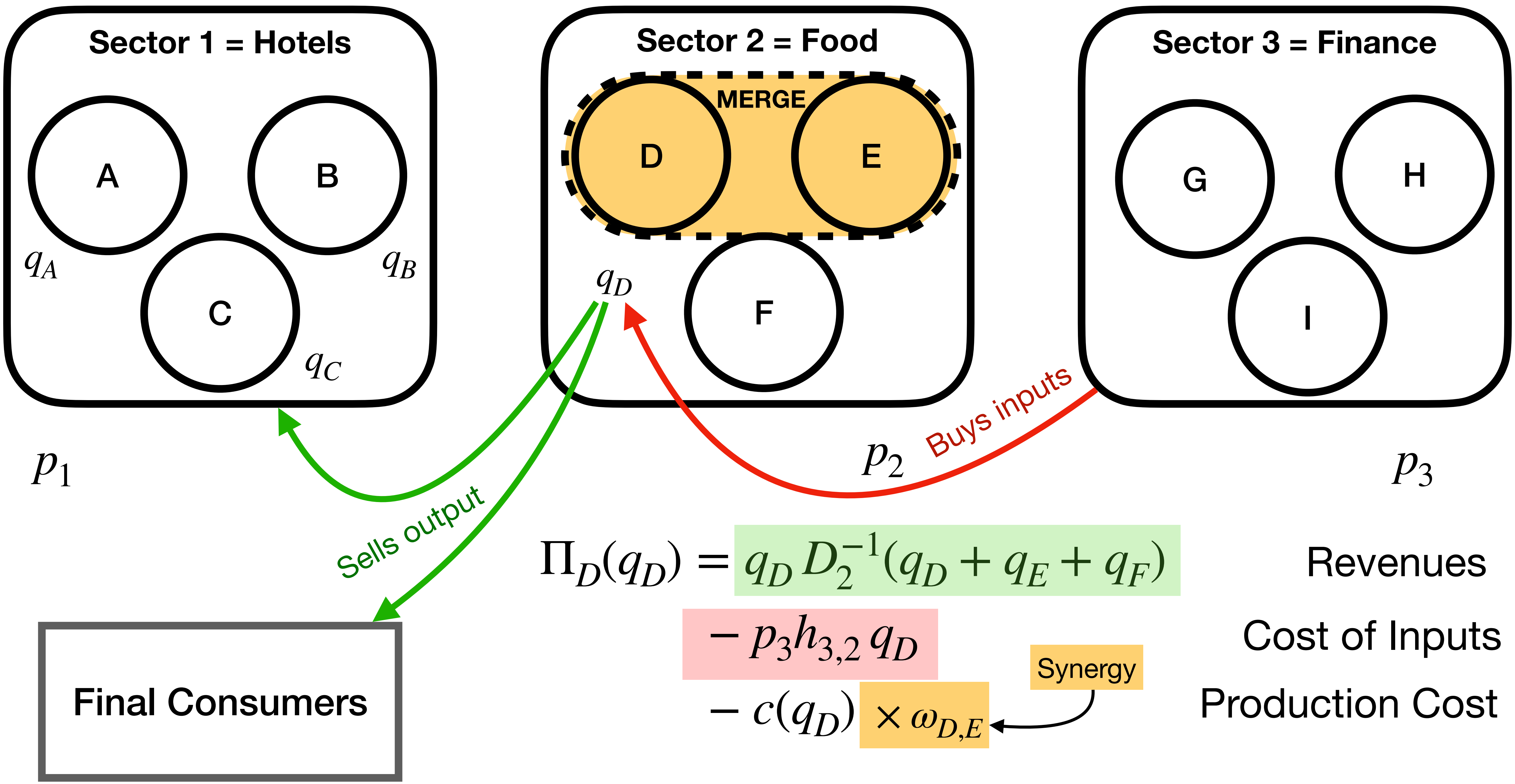

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$$- \, p_3 h_{3,2}\, q_D \quad \text{Cost of Inputs}$$

$$- \, c(q_D) \times \omega_{D,E} \quad \text{Production Cost}$$

Synergy

Theoretical Model (omitted + simplified)

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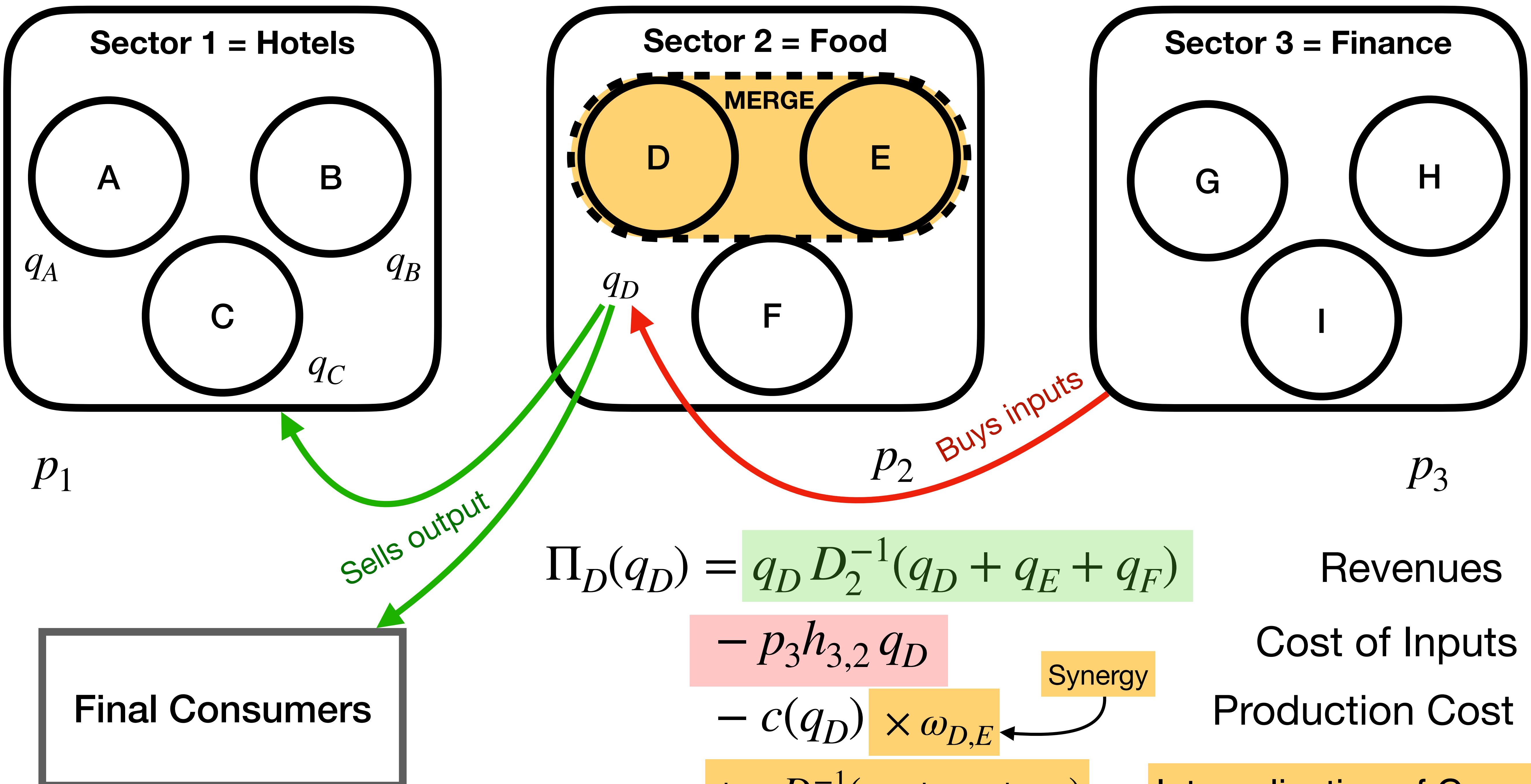

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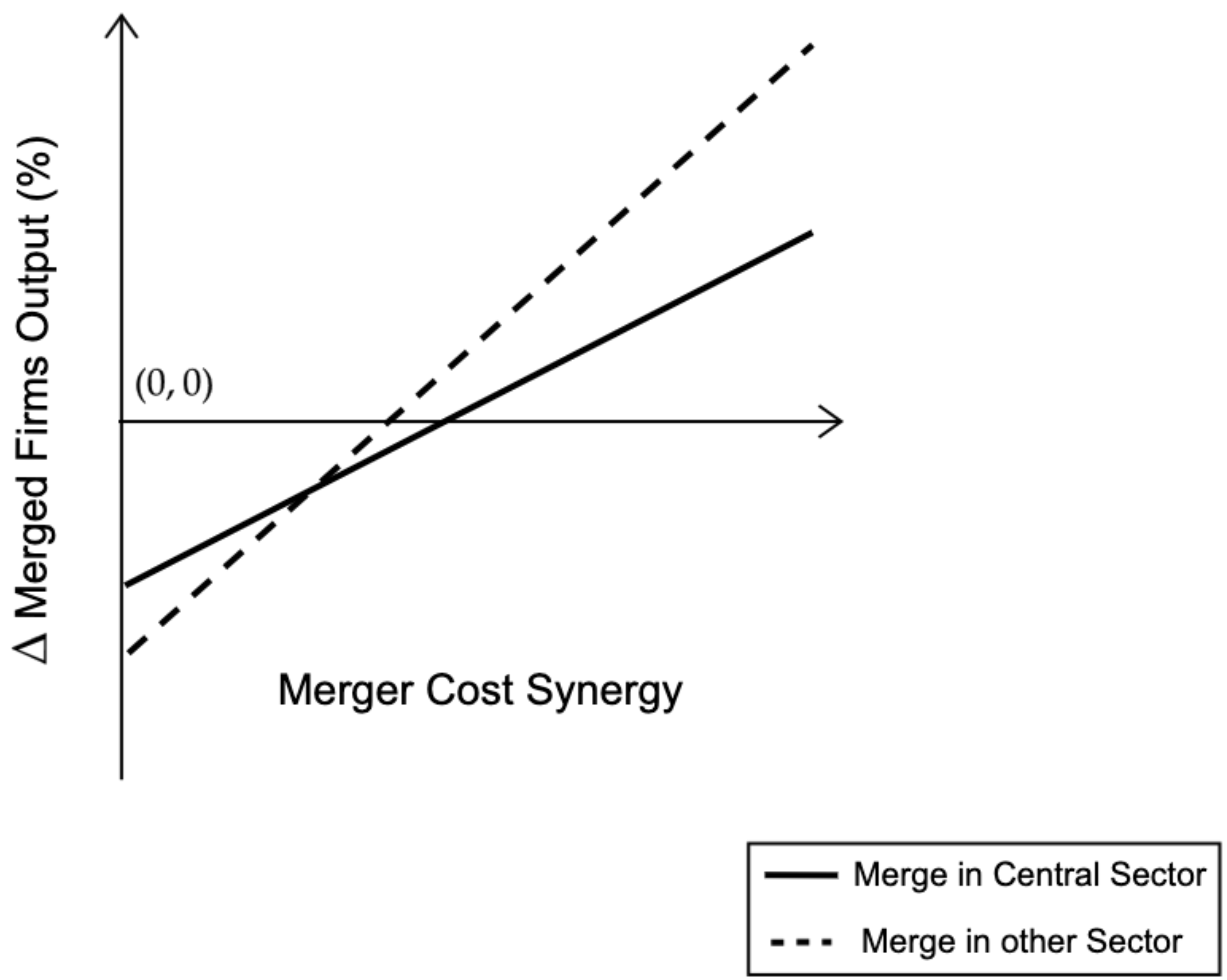
$$+ q_E D_2^{-1}(q_D + q_E + q_F)$$ Internalization of Comp

Theoretical Results

Following a merger, in equilibrium, quantities…



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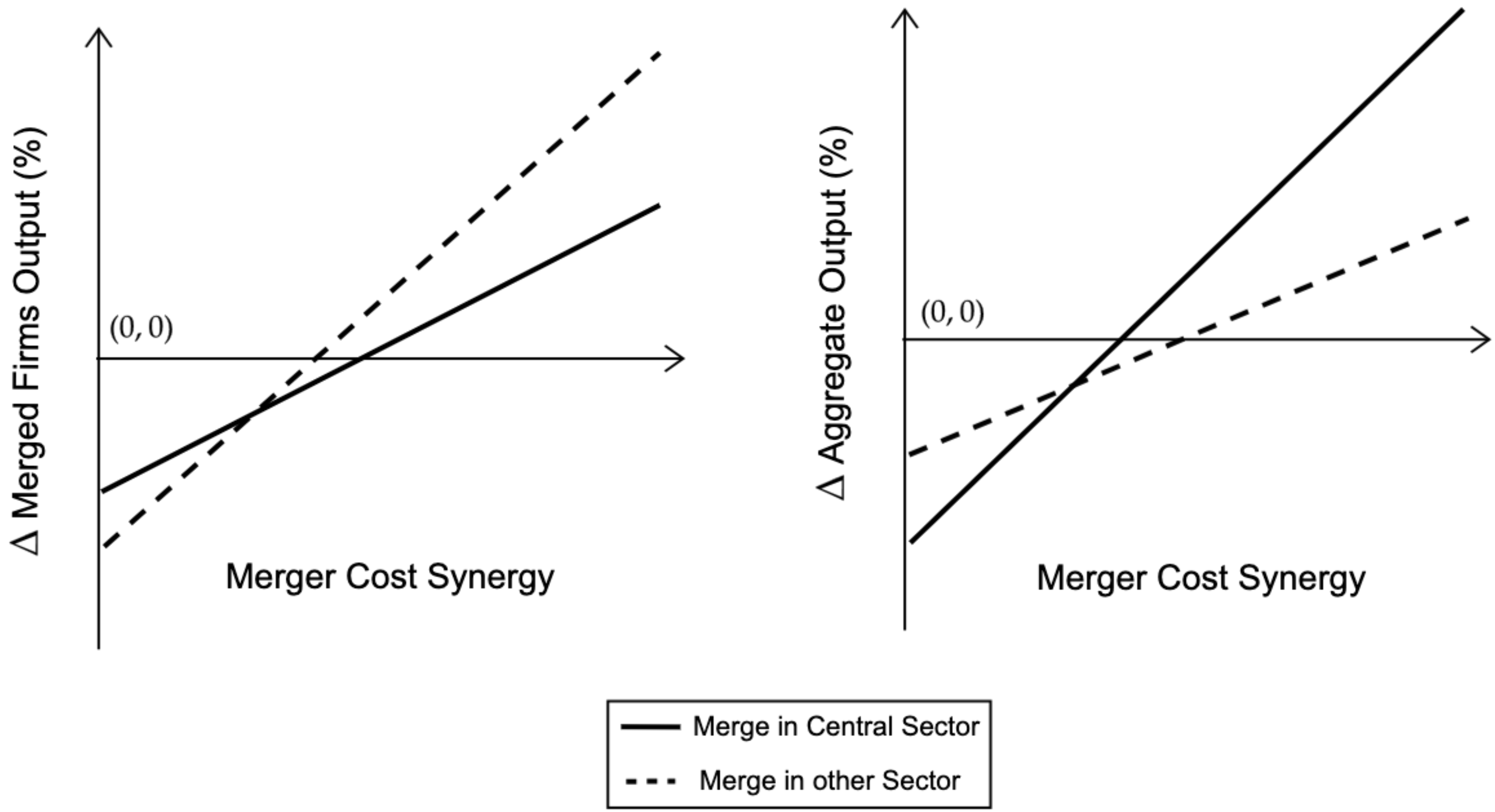

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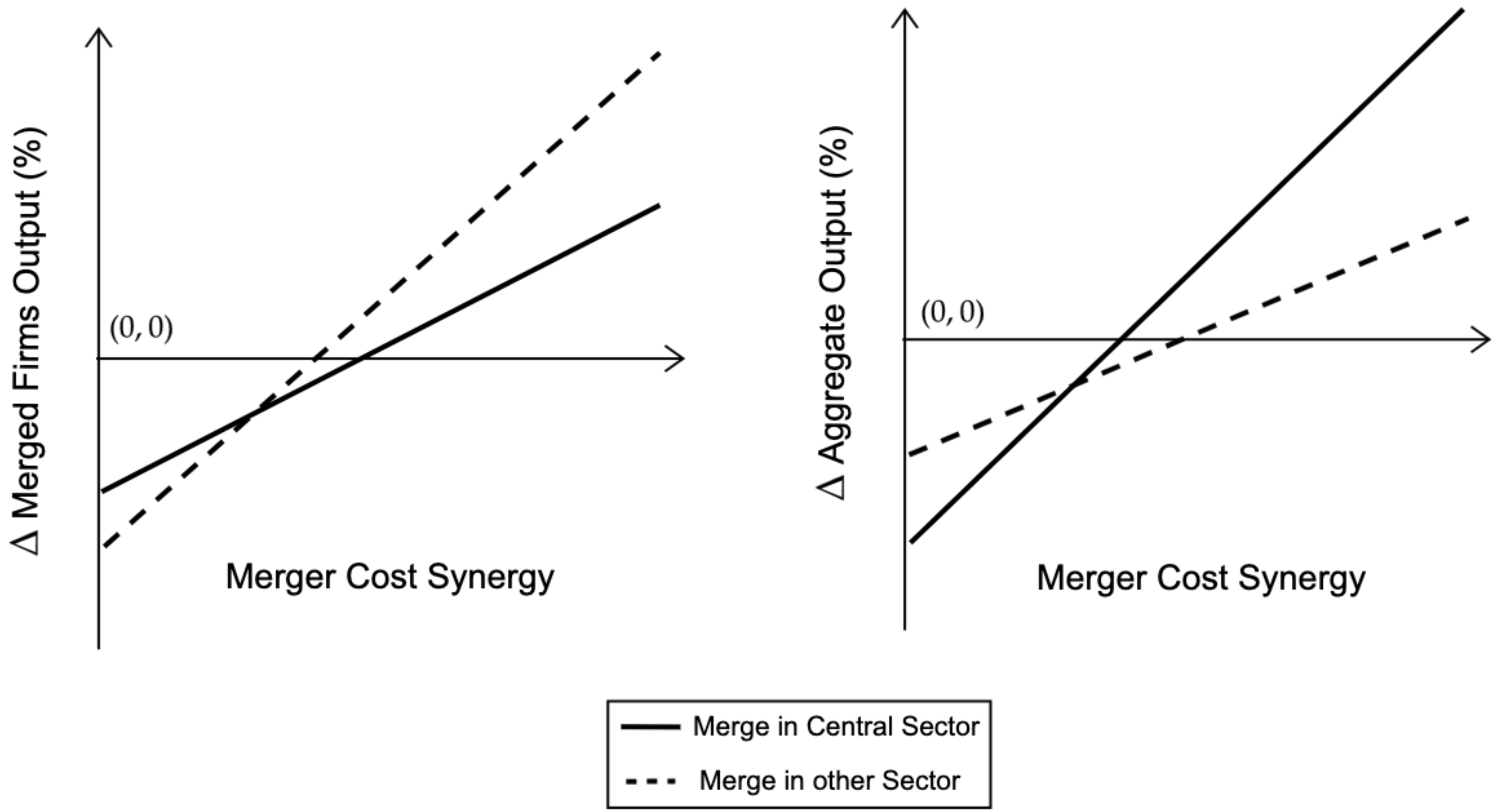
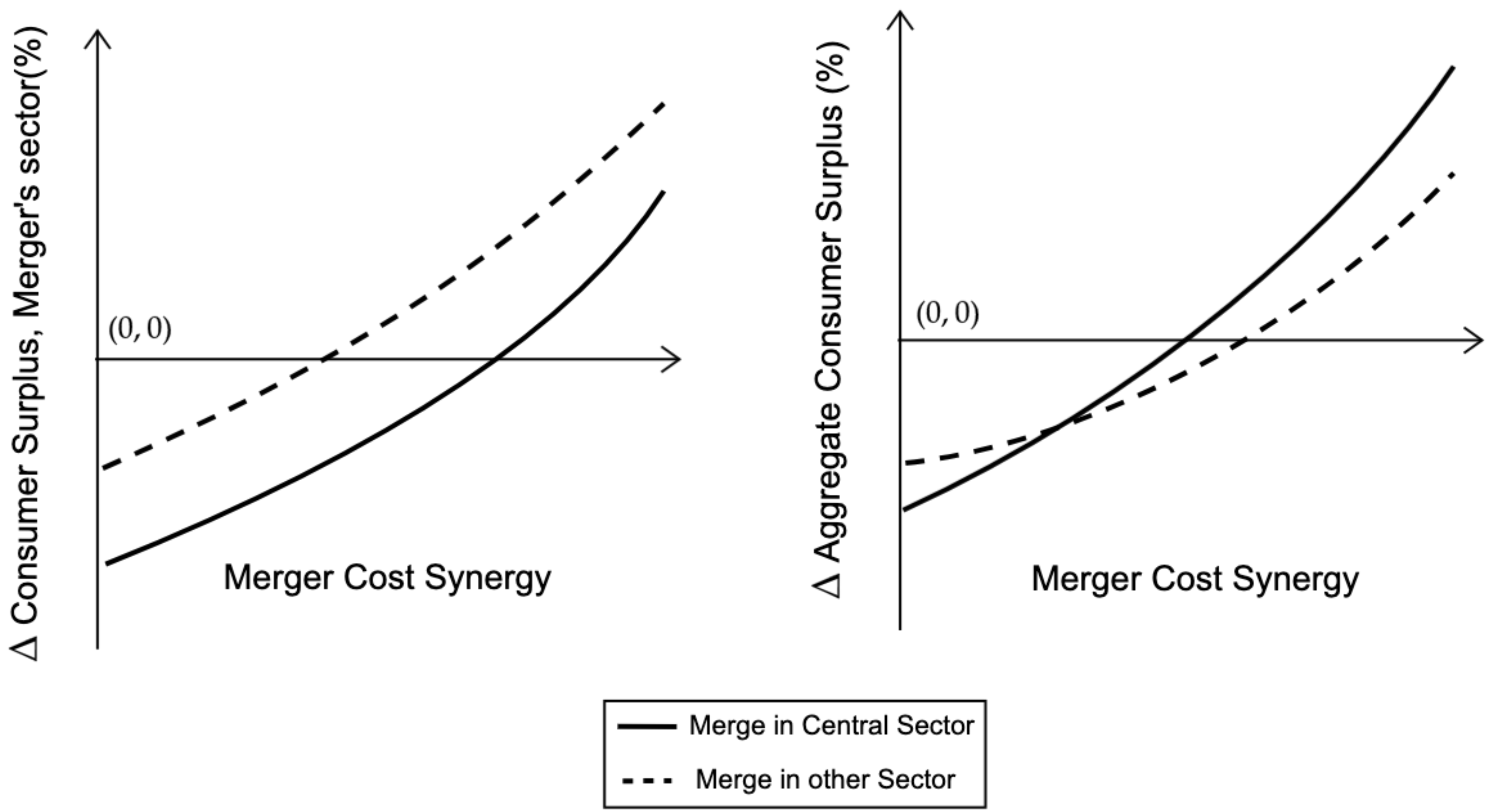


Key idea: Merger in a central sector → less incentives to change quantities

… but larger aggregate effects

And (final) Consumer Surplus…



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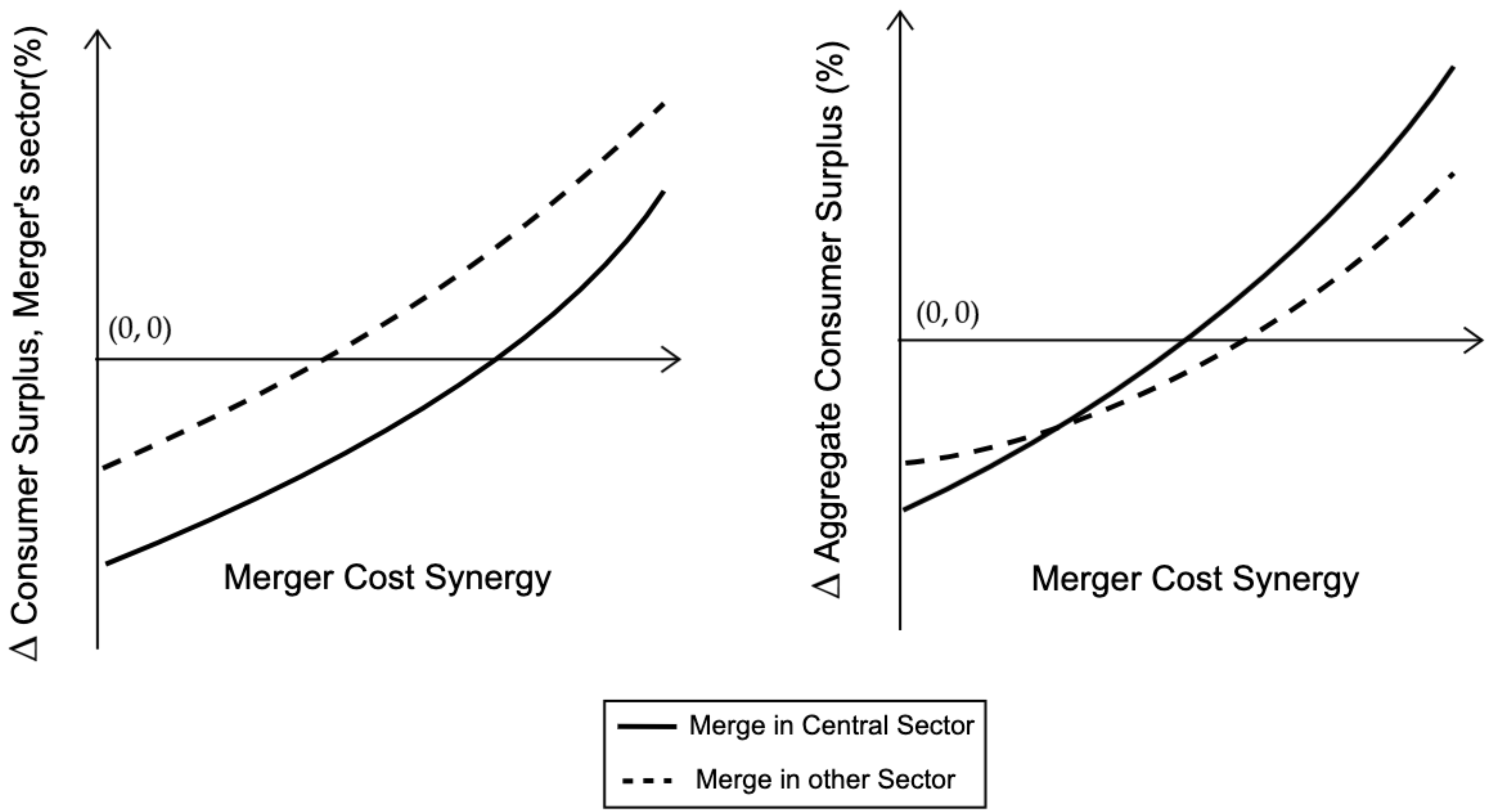

Key ideas: - CS merger sector ≠ Aggregate CS

- Lower synergies needed for a central sector merger to increase CS

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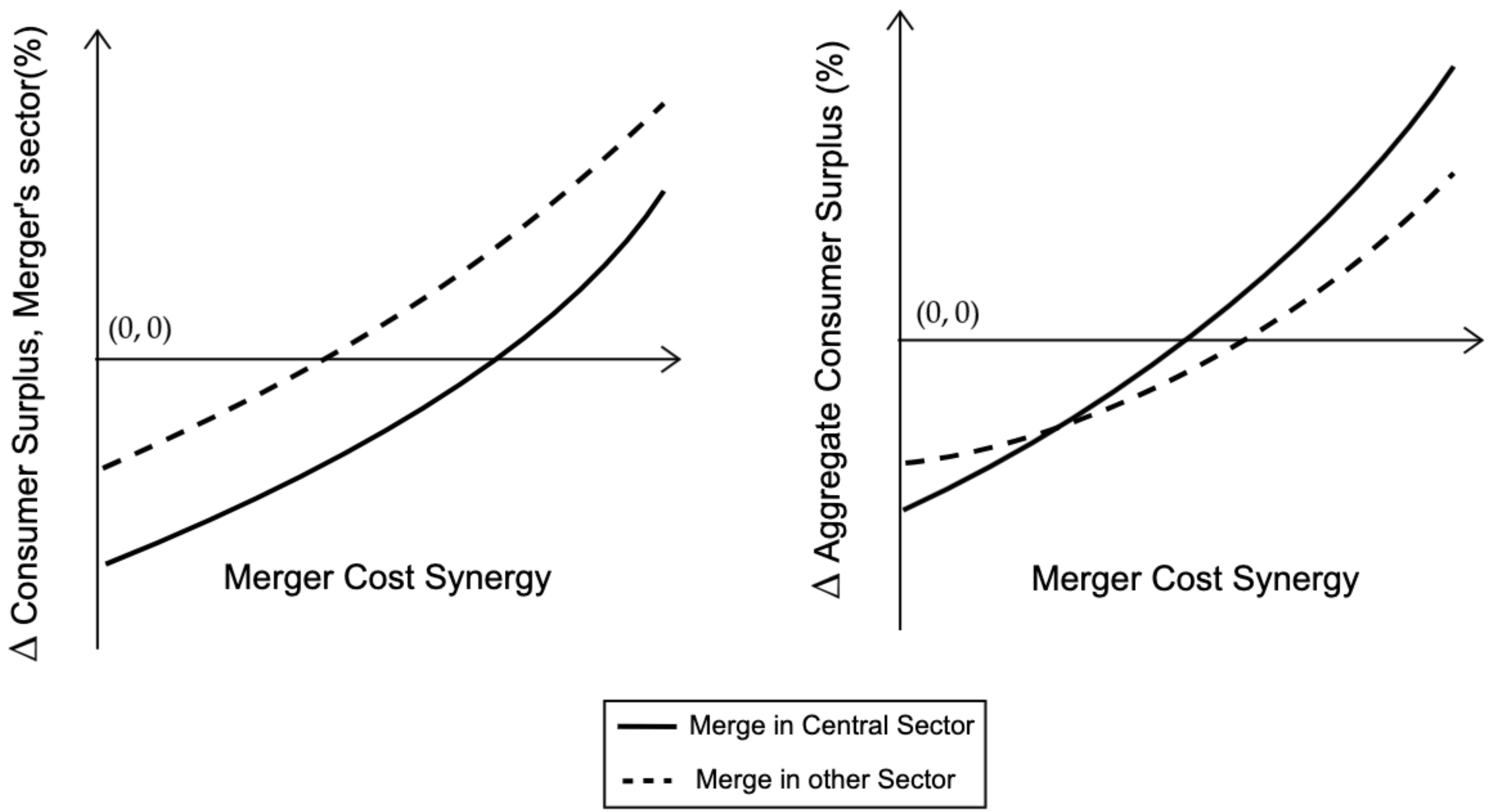

Key ideas: - CS merger sector ≠ Aggregate CS **(But riskier!)**
- Lower synergies needed for a central sector merger to increase CS

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Does the decision depend on the centrality of the sector?

Empirical result:

Sum of pre-merger market shares

$$\boxed{\mathbb{P}(\text{prohibited}\,|_{\text{phase 2}}) = \beta_0 + \beta_1\text{Centrality} + \beta_2\text{HHI} + \beta_3\text{TotalMS} + \beta_4\text{TotalMS} \times \text{HHI} + \varepsilon_i}$$

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(intercept)	0.1242	0.044	0.005
Centrality	**-0.6400**	**0.062**	**0.000**
HHI	0.0108	0.013	0.399
TotalMS	-0.0026	0.001	0.003
HHI×TotalMS	0.0008	0.000	0.000
R^2		0.373	
N. obs.		1264	

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Empirical result:

Sum of pre-merger market shares

$$\mathbb{P}(\text{prohibited}\,|_{\text{phase 2}}) = \beta_0 + \boxed{\beta_1}\text{Centrality} + \beta_2\text{HHI} + \beta_3\text{TotalMS} + \beta_4\text{TotalMS} \times \text{HHI} + \varepsilon_i$$

Variable	**Coefficient**	**Std. Error**	**p-value**
(intercept)	0.1242	0.044	0.005
Centrality	**-0.6400**	**0.062**	**0.000**
HHI	0.0108	0.013	0.399
TotalMS	-0.0026	0.001	0.003
HHI×TotalMS	0.0008	0.000	0.000
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Final consumer protection?

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What am I missing here? → ivan.rendo@tse-fr.eu